

Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring ^{*}

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Abstract

This paper provides new evidence on the multitask pressure local governments face in allocating land between agricultural production and urban-industrial development. I examine a tightening of cropland protection supervision in China, which strengthened enforcement of area targets in bureaucratic evaluations while leaving land quality weakly monitored. Using grid-level measures constructed from satellite imagery to track cropland changes, I implement a difference-in-differences design that exploits cross-county variation in cropland protection pressure. Within each prefecture, compensation obligations are assigned across counties according to each county's predetermined share of the prefecture's available rural-land reservoir. I find that high-pressure counties respond by expanding cropland after the reform, but the newly reclaimed plots are less agronomically suitable and are used less intensively, exhibiting lower vegetation cover, persistently higher fallow rates, and lower crop output per hectare. A spatial regression-discontinuity design comparing plots across administrative borders separating jurisdictions with sharply different cropland protection pressure supports the same finding. Mechanism evidence indicates that the decline in reclaimed-land quality reflects distortions in local governments' reclamation choices, rather than only movement down the land-quality frontier: reclamation shifts toward less-monitored and lower-cost sites, and quality losses are larger when officials face stronger career incentives or greater economic-growth pressure. Finally, additional evidence shows that counties absorbing larger compensation burdens later supply less construction land, suggesting that market-based trading may mitigate cropland quality and productivity losses but also exacerbate spatial inequality by limiting the future development potential of land-abundant counties that assume greater cropland protection responsibilities.

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1 Introduction

Fertile and accessible locations have long supported both agricultural production and dense settlement, leaving many cities surrounded by highly productive cropland. Yet this spatial overlap increasingly exposes cropland to competition from urban development. Bren d'Amour et al. (2017) estimate that urban expansion between 2000 and 2030 will convert 27 to 35 million hectares of cropland worldwide and that the affected land is 1.77 times as productive as the global average. For China, the corresponding estimate is 7.1 to 8.6 million hectares, equivalent to 8.2 to 9.8 percent of national crop production. Governments therefore face a central policy design problem: how to balance the land demands of urban development with the need to maintain cropland and sustain agricultural production.

China has responded to these pressures by strengthening cropland protection. Political authority in China is highly centralized, but policy implementation is decentralized to local governments, which retain substantial discretion over how central mandates are carried out (Xu, 2011; Zhang et al., 2018). Local bureaucrats implement multiple policy mandates and thus operate as multitasking agents (Holmstrom and Milgrom, 1991; Chen et al., 2018). Because implementation is decentralized and information is asymmetric between the central and local governments, the center cannot fully observe how these mandates are carried out. A natural response is to strengthen monitoring and performance-based assessments. Yet while monitoring remains incomplete within a task, with some dimensions observable and others not, local governments may concentrate compliance on the dimensions receiving greater scrutiny and shift distortions toward those that remain less observable.

This paper asks: when the center tightens supervision over one dimension of the cropland-protection task, does local implementation shift distortions toward less closely monitored dimensions of the same task? Cropland protection in China places local governments under two linked incentive tensions. The first is a multitask tension between urban expansion and cropland protection: local governments support urban development by converting cropland to construction use while remaining responsible for maintaining cropland quantity and quality. The second lies within cropland protection itself: the central government seeks to safeguard both cropland area and productive capacity, treating area as the basis and productive capacity as the core. I examine whether, when the conflict between urban expansion and cropland protection becomes more acute, compliance with the more closely monitored area requirement generates losses in less closely monitored land quality and actual cultivation.

Local governments in China have long been evaluated primarily on economic performance, particularly economic growth (Fan et al., 2025). To support industrialization and urban expansion, local officials have strong incentives to convert land for construction use, making land allocation a central instrument of local development policy. At the same time, concerns over food security led the central government to adopt the *Cropland Occupation–Compensation Balance Policy*, which requires each unit of cropland converted to construction use to be offset by an equal area of newly reclaimed cropland. Local governments thus operate in a multitask environment that requires balancing economic development with cropland protection.

Prior to 2017, however, cropland protection was weakly enforced despite the formal existence of quantitative targets. This changed with the 2017 reform of the cropland protection responsibility assessment system, under which compliance with cropland protection targets became a binding component of official evaluations and promotion criteria. While the reform aimed to strengthen the joint management of cropland quantity and productive capacity, anecdotal evidence suggests that implementation gave greater emphasis to the readily observable quantity target, with cropland quality and subsequent use remaining less closely monitored. This asymmetry raises a key concern: when quantitative targets become binding under incomplete monitoring, local governments may comply by expanding cropland area through the reclamation of lower-quality plots, generating distortions in land allocation.

In this paper, I use satellite-based raster data to identify cropland and cultivation intensity. The unit of observation is a 500×500 meter grid cell. Aggregating raster measures to the grid level allows me to track changes in land allocation and land use at a spatial scale relevant for local policy implementation, while avoiding reliance on administrative statistics that may be subject to reporting biases.

I leverage cross-county variation in cropland protection pressure to identify the effects of the 2017 enforcement tightening on land allocation and cultivation outcomes. Cropland protection pressure captures a county's predicted cropland-compensation burden relative to its available rural land reservoir. Under the *Cropland Occupation-Compensation Balance Policy*, compensation obligations generated by core urban districts can be met by non-core counties within the same prefecture. The prefecture-level obligation is assigned from above, while its allocation across counties depends on each county's predetermined share of the prefecture's available rural land. Because these shares reflect pre-policy land endowments and administrative boundaries rather than contemporaneous county policy choices, the resulting cross-county variation in exposure is plausibly exogenous to local responses to the reform. I therefore use cropland protection pressure as a continuous measure of exposure to the common 2017 enforcement shock and compare changes before and after the reform across counties with different predetermined exposure in a difference-in-differences design.

The main difference-in-differences estimates reveal a quantity-quality trade-off in local policy implementation. Cropland shares followed similar trends before 2017 but rose more thereafter in counties facing greater cropland protection pressure. Yet newly reclaimed plots in these counties are less suitable for cultivation, more likely to remain fallow, and exhibit lower vegetation coverage after reclamation. Tighter enforcement therefore increases the amount of land recorded as cropland while drawing less suitable and less intensively used plots into the cropland inventory.

A spatial regression-discontinuity design across provincial borders yields the same pattern under a different source of identifying variation. Along the provincial border, cropland expansion and fallowing both increase on the higher-pressure side, while pre-policy changes remain smooth at the border. This border design corroborates the extensive-margin expansion together with higher fallow and lower vegetation coverage found in the county-level analysis.

Additional evidence shows that post-reform reclamation in high-pressure counties moves farther from existing cropland and toward more isolated locations. The evidence does not support normal seasonality, temporary land preparation, or crop switching as explanations for the utilization results: fallowing rises during the growing season, persists rather than dissipates over time, and is not accompanied by clear shifts in crop composition or cropping frequency.

The mechanism analysis points to multitask incentives operating under incomplete monitoring. The decline in the suitability of reclaimed land is larger when prefecture leaders face stronger promotion incentives. When pressure to sustain economic growth is greater, local governments expand cropland while continuing to allocate land to urban construction, and the additional cropland is less suitable. Reclamation also shifts away from provincial quality-monitoring stations after 2017. These patterns are consistent with multitask incentives shifting implementation toward dimensions of the mandate that remain less observable.

This paper contributes to the analysis of a central land-use conflict: balancing urban expansion with the need to protect cropland and sustain agricultural production. Yu (2026) also studies China’s cropland protection policy and is closely related to this paper. Its analysis focuses on the extensive margin, examining how restrictions on cropland conversion to urban use constrain urban land supply and generate aggregate welfare losses. This paper extends the analysis of cropland protection to the intensive margin by studying the 2017 tightening of enforcement. The reform strengthened enforcement of cropland quantity while leaving quality and actual use less closely monitored. I show that this uneven tightening shifts distortions toward less-monitored margins: local governments meet the observable area target by reclaiming lower-quality land that is cultivated less intensively.

Beyond this, the paper also speaks to three broader strands of the literature. First, it connects to the principal–agent literature on performance measurement and multitasking (Holmstrom and Milgrom, 1991; Baker, 1992; Dixit, 2002; Courty and Marschke, 2004; Bandiera et al., 2021). A growing empirical literature examines how governments respond to multiple, imperfectly measured, or competing objectives (Chen et al., 2018; He et al., 2020; Cao et al., 2026; Hong and Huang, 2025; Fan et al., 2025; Fang et al., 2023; Axbard and Deng, 2024; Fisman et al., 2025). Much of this work centers on effort allocation across distinct tasks. This paper extends that literature by documenting distortions across unequally observable dimensions of a single policy mandate.

Second, the paper speaks to the literature on agricultural policy and land-use regulation (Cai et al., 2017; Henderson et al., 2022; Fu et al., 2021; Kong et al., 2024; Fang et al., 2026; Lu et al., 2026; Hong et al., 2025; Chen and Li, 2025; Xing et al., 2026; Restuccia and Santaella-Llopis, 2017; Adamopoulos and Restuccia, 2014; Chari et al., 2021; Chen et al., 2022, 2023; Adamopoulos et al., 2024). Adamopoulos et al. (2022) study how the allocation of agricultural inputs affects productivity. This paper focuses particularly on land and on a distinctive institutional setting in which local governments direct its allocation across uses. They determine which parcels enter agricultural production and which are assigned to urban construction or other uses. In this setting, governments can influence agricultural output through their

allocation of land to agricultural production.

Finally, this paper speaks to the literature on the Chinese economy, particularly research on decentralization and local incentives in policy implementation (Montinola et al., 1995; Jin et al., 2005; Zhang, 2006; Xu, 2011; Li and Zhou, 2005; Zhang et al., 2018; Wang et al., 2020; Wang and Yang, 2025; Luo et al., 2025). This literature shows how decentralized implementation, top-down planning, and performance-based mandates shape local government behavior, with particular attention to information frictions and strategic responses. This paper enriches the literature by showing how institutional design itself can generate adverse effects even without collusion or deliberate manipulation. When monitoring is uneven and local governments face binding resource constraints, compliance with primary, observable targets can come at the expense of less observable environmental and agricultural outcomes.

The remainder of the paper is organized as follows. Section 2 describes the institutional background and China’s cropland protection policy. Section 3 presents the data and measurement of cropland protection pressure. Section 4 documents the motivating facts. Section 5 presents the empirical strategy and main results. Section 6 examines the mechanisms underlying the land-use responses. Section 7 quantifies their economic magnitudes and incidence. Section 8 discusses the findings and concludes.

2 Institutional Background

2.1 Cropland Occupation–Compensation Balance Policy

To ensure food security and agricultural self-sufficiency, the Chinese government has long treated cropland protection as a national policy priority. China’s modern cropland-protection framework dates to the 1990s, when rapid industrialization and urban expansion made limiting the loss of agricultural land increasingly urgent (State Council of the People’s Republic of China, 1998).¹

The principal policy instrument linking urban expansion to cropland protection is the *Cropland Occupation–Compensation Balance Policy*. Introduced nationally in the late 1990s, the policy imposes a one-for-one replacement rule: when a local government converts cropland to construction or another non-agricultural use, it must create an equal area of compensation cropland elsewhere. The rule therefore converts each unit of planned cropland occupation into an equal-area compensation obligation.

Implementation operates through China’s hierarchical land-use planning system. Upper-level governments assign a *planned land-use quota* to each lower-level jurisdiction, which caps the amount of agricultural land that may be converted into new construction land. The quota is therefore a land-conversion constraint rather than a construction-land-supply target. Under the one-for-one rule, the amount of cropland planned for conversion under this quota simultaneously

¹The aggregate cropland-retention target within this broader framework is commonly referred to as the “cropland red line.”

determines the jurisdiction’s cropland-compensation obligation. The allocation cascades from the central government to provinces, prefectures, and counties.²

Local governments have limited spatial flexibility in meeting the resulting obligation. If a county cannot create sufficient compensation cropland within its own jurisdiction, compensation may be arranged across counties within the same prefecture. If prefecture-level adjustment is infeasible, the obligation may be met across prefectures within the same province, typically through a paid transfer mechanism. These arrangements separate the jurisdiction that generates a compensation obligation from the jurisdiction that supplies the replacement cropland, a feature central to the empirical design.

2.2 Policy Revision and Monitoring Tightening in 2017

Despite formal protection rules, cropland area continued to decline and only became relatively stable after 2017 (Appendix Figure B.1, Panel A). Both the pre-2017 decline and the post-2017 rebound are more pronounced among the six province-level units with the highest GDP per capita in 2021 (Panel B).

To strengthen enforcement of the Cropland Occupation–Compensation Balance Policy, the State Council issued the *Opinions on Strengthening Cropland Protection and Improving the Balance of Cropland Occupation and Compensation* in January 2017 (hereafter, the 2017 Opinions) (State Council of the People’s Republic of China, 2017). The Opinions retained the policy’s one-for-one structure but made principal local leaders accountable and linked cropland-protection performance to cadre and ecological civilization assessments. Revised provincial assessment methods followed in January 2018; provincial and prefectural governments then adopted corresponding measures for lower-level jurisdictions. Appendix Figure B.2 summarizes this administrative rollout.

The central government framed the revision as protecting both cropland quantity and productive capacity. In practice, however, contemporaneous accounts and government reports indicate that the quantitative cropland-area target was systematically assessed, while the quality of compensation cropland received much less scrutiny (see Appendix Section D.2). Writing in *People’s Daily*, Wang (2023) documents low-quality reclamation and risks of abandonment and subsequent loss of newly created cropland.

2.3 Local Bureaucrats and Urban Expansion

China’s local officials implement multiple policy objectives under a performance-based personnel system. Promoting economic growth is their primary task. The evaluation system also encompasses resource use and environmental protection. Cropland protection forms part of this broader evaluation framework.

²Under China’s administrative hierarchy, county-level jurisdictions include counties, county-level cities, autonomous counties, and municipal districts. These units occupy the same formal administrative tier; the paper therefore uses “county-level” as an institutional classification rather than a sample-specific convention.

Land conversion links these objectives. Additional construction land accommodates manufacturing, infrastructure, and urban development. Conveying state-owned construction land also generates fiscal revenue. Official fiscal rules require land-conveyance receipts to be paid into local treasuries and budgeted through local government-fund accounts; permitted uses include land development and urban infrastructure (see Appendix Section D.1). These production and fiscal channels link urban expansion to local economic performance and sharpen the conflict with cropland protection.

2.4 Government-Led Land-Use Allocation

China's land-use system is organized around legally defined use controls. The Land Administration Law classifies land as agricultural, construction, or unused, requires land-use planning, and restricts the conversion of agricultural land to construction use through an approval process (see Appendix Section D.1). Transactions therefore allocate use rights only within the legally and administratively permitted category. The institutional sequence differs across the two categories.

For construction land, city or county authorities announce the location, permitted use, planning conditions, and conveyance terms of a state-owned parcel and assign its land-use right through invitation to bid, auction, or public listing; the winning user then signs a fixed-term conveyance contract. Agricultural land, by contrast, generally remains collectively owned and is contracted to rural households. Once a parcel, including reclaimed land, is designated for agricultural use, the contractor may cultivate it directly or transfer the operating right through lease or other legally permitted arrangements, but the transaction may not change its agricultural use. Thus, both construction and agricultural transactions occur within an administratively determined use category, while the relevant market instrument differs: land-use-right conveyance for construction land and transfer of operating rights for cropland (see Appendix Section D.1).

2.5 Jiangsu Province

The analysis focuses on Jiangsu because the cropland-compensation constraint is not uniformly binding across China. Appendix Figure B.3 maps provincial compensation obligations per unit of land area and shows that high-intensity obligations are concentrated in the eastern coastal region. Official reports and policy documents from Jiangsu, Zhejiang, and Guangdong explicitly state that these provinces face acute land-use conflicts and difficulty meeting cropland-compensation obligations locally (see Appendix Section D.3). By contrast, in provinces with slower urbanization, weaker growth pressure, persistent population outflows, or more abundant reserve land, local governments face a more limited trade-off between urban expansion and cropland retention. A nationwide analysis would therefore combine jurisdictions in which the constraint is binding with many in which it is not, obscuring the mechanism of interest.

Jiangsu provides a sharp setting for studying a binding cropland-compensation constraint.

Located in the Yangtze River Delta, it is both a major agricultural province, with a long history of intensive cultivation, and one of China’s most economically developed provinces, with high GDP per capita and rapid urbanization. Its predominantly flat terrain makes almost all land physically suitable for cultivation, while sustained industrialization and urban expansion generate strong and persistent demand for construction land. Local officials therefore face an acute conflict between economic development and cropland protection. Rather than estimating nationwide aggregate effects, the paper asks whether a binding compensation obligation leads local governments to reclaim land that is less suitable or cultivated less intensively. Taken together, these considerations motivate the use of Jiangsu data.

3 Data and Measurement

3.1 Data Sources and Measurement

This study primarily relies on satellite-based raster data to measure land characteristics at a fine spatial resolution. To harmonize datasets with different spatial and temporal resolutions and to facilitate empirical analysis, I overlay all raster layers onto a uniform 500×500 meter grid covering the study area. For each grid cell, I compute average values of the underlying raster measures, which constitute the main unit of observation in the analysis. Table 1 summarizes the principal data sources. Appendix Table C.1 reports the additional data sources.

3.1.1 Cropland

As summarized in Table 1, cropland information is obtained from the 30-meter annual land cover dataset for China (CLCD; Yang and Huang (2021)). I use the CLCD land-cover class labeled “cropland” as the empirical measure of cropland throughout the analysis and use annual files from 2012 through 2022. The dataset distinguishes among cropland, forest, shrubland, grassland, barren land, impervious surface, water bodies, snow, and ice. It therefore allows me to identify both cropland and built-up areas, with the latter captured by the impervious-surface category.

3.2 Fractional Vegetation Cover and Measurement of Fallow Rate

Fractional Vegetation Cover (FVC) data are obtained from the National Tibetan Plateau Data Center (Zhao et al., 2023). The original dataset provides spatiotemporally continuous vegetation cover for China at a 30-meter resolution and a bimonthly frequency from 2010 to 2022. Following standard practice in the remote sensing literature, I aggregate the bimonthly observations to annual averages at the grid-cell level, which serve as a measure of vegetation coverage for each grid-year.

I use FVC to measure land-use intensity and to define fallow land. A grid cell is classified as fallow in a given year if its annual average FVC is below 0.2, which corresponds to the bottom

25 percent of the FVC distribution across cropland grids. This threshold captures land with very low vegetation cover and limited cultivation activity. In Appendix Figure B.4c, I examine the sensitivity of the results to alternative FVC cutoffs. The estimates remain stable when the threshold is chosen sufficiently low, indicating that the main findings are not driven by the specific choice of cutoff.

3.3 Soil Type and Measurement of Soil Suitability

The first soil measure comes from the 1:1,000,000 Soil Map of China compiled by the National Soil Survey Office in 1995. I construct a second measure from the Harmonized World Soil Database v2.0, released in 2023 at an approximately 1-kilometer resolution. For both sources, soil units are mapped to an analogous five-category suitability scale and aggregated to the analysis grid. The agreement between the two independently constructed classifications reduces dependence on any single soil map. A potential limitation of these measures is that soil characteristics are time-invariant over the sample period. However, this feature is unlikely to pose a major concern for the analysis, since soil type reflects long-run geological and pedological conditions that evolve slowly over time and are largely predetermined with respect to recent policy changes.

I measure land quality using a grid-level soil suitability index constructed from soil type classifications as shown in Appendix Table C.2. The mapping from soil types to suitability scores follows the FAO Land Evaluation Framework and China’s National Land Capability Standard (GB/T 28407–2012), which provides qualitative guidelines for assessing agricultural suitability based on soil properties. Because these frameworks do not offer a one-to-one correspondence between specific soil types and numerical suitability scores, the resulting index should be interpreted as a coarse proxy for inherent land quality rather than a precise agromonic measure.

Specifically, I assign each soil type a suitability score ranging from 1 to 5, corresponding to FAO suitability classes from permanently unsuitable to highly suitable, and compute the grid-level score as the average across soil types within each 500×500 meter grid cell. To facilitate interpretation and reduce sensitivity to the exact scoring scheme, I additionally classify land as suitable if the average score is at least 3, and unsuitable otherwise. This cutoff separates soils that are marginally suitable or better from soils that are saline, alkaline, waterlogged, or otherwise non-arable, which correspond to the lowest suitability classes.

3.3.1 Administrative and Socioeconomic Data

County-level socioeconomic variables are obtained from the China County Statistical Yearbooks. The data include information on economic activity, agricultural production, population, and fiscal conditions. Cropland compensation obligations and the classification of core urban districts are collected from prefecture-level Land Use Master Plans, which specify each jurisdiction’s targets for urban expansion and cropland compensation through 2020. Administrative

boundaries come from mainland China county-level shapefiles distributed through Harvard Dataverse. Appendix Table C.1 summarizes the remaining data sources.

3.4 Measuring Cropland Protection Pressure

This subsection constructs a county-level measure of exposure to the prefecture-level cropland compensation burden. The measure captures the cropland protection pressure faced by local officials, which reflects the intensity of multitask pressure arising from the need to satisfy cropland protection targets while accommodating urbanization and development demands. It serves as the key explanatory variable in the empirical analysis.

The construction of the cropland protection pressure measure exploits institutional features of China’s Cropland Occupation–Compensation Balance Policy. Two policy rules are central to this design:

1. **One-for-one occupation–compensation rule.** Any conversion of cropland to construction land must be offset by the creation of an equivalent amount of compensation cropland.
2. **Non-local compensation with prefecture-level priority.** Compensation is not necessarily required to occur within the converting county. When core urban districts lack sufficient land, cropland compensation is, in principle, fulfilled through *cross-county adjustment within the same prefecture*. Only if prefecture-level adjustment is infeasible may compensation be arranged across prefectures within the same province, typically through a paid transaction mechanism.

Together, these rules imply that cropland compensation is organized around prefecture-level obligations and the non-core available-land reservoir. Because core urban districts are primarily tasked with meeting economic growth targets, their cropland compensation obligations are frequently met by counties within the same prefecture.³ As a result, non-core counties absorb compensation obligations generated by urban expansion, creating variation in cropland protection pressure across counties.⁴

Based on this structure, I define cropland protection pressure at the county level as

$$\text{Pressure}_c = \left(\frac{\text{Obligation}_p}{\text{TotalLand}_p} \right) \times \text{LandShare}_c, \quad (1)$$

where p denotes the prefecture to which county c belongs. Obligation_p measures the total cropland compensation obligation assigned to the core urban districts of prefecture p under

³This practice is documented in official planning materials. For example, Lianyungang City’s Land Use Master Plan states that “districts such as Haizhou District and Lianyun District that are unable to achieve local cropland reclamation–replacement balance due to insufficient land reserves may, upon approval, implement off-site reclamation–replacement within the municipal jurisdiction, provided that the aggregate reclamation obligation remains unchanged.”

⁴The definition of the core urban area is specified in each prefecture’s Land Use Master Plan (2006–2020).

provincial land-use planning targets. TotalLand_p captures the total area of available land outside core urban districts within prefecture p , excluding impervious surfaces and water bodies, and represents the effective land reservoir for cropland compensation. LandShare_c denotes county c 's share of this reservoir, defined as the ratio of available land in county c to the total available land summed across all non-core counties within the same prefecture.

This construction implies that counties with larger shares of non-core, available land are more exposed to cropland compensation pressure when prefecture-level obligations increase. Importantly, LandShare_c is determined by pre-existing land endowments and administrative boundaries, and is fixed prior to the tightening of cropland protection in 2017. As a result, conditional on prefecture-level obligations, cross-county variation in cropland protection pressure reflects an exogenous allocation of the delegated compensation burden rather than endogenous policy choices by county governments. Conceptually, this structure resembles a shift-share design, in which prefecture-level compensation obligations act as the common shock, while county-level land shares determine differential exposure. This feature underpins the empirical strategy used in subsequent sections.

Figure 1 maps the standardized cropland protection pressure index across Jiangsu counties, while Appendix Figure B.5 summarizes the distribution of the unstandardized index across counties by prefecture. The map shows substantial spatial heterogeneity in exposure. The box plot shows both differences in prefecture-level medians and dispersion across non-core counties within the same prefecture. Together, the two figures show that the pressure index has sufficient variation both across and within prefectures.

To formally distinguish the within- and between-prefecture components of variation in this key variable, I conduct a simple variance decomposition based on the ANOVA identity.⁵ Specifically, total variation in Pressure_c is decomposed into variation across prefecture-level means and variation across counties within the same prefecture. The results indicate that 39.8% of the total variation in the pressure index arises from within-prefecture differences across counties, while the remaining 60.2% reflects differences across prefectures.

4 Motivating Facts

4.1 Land-Use Area Trends by Cropland Protection Pressure

Descriptive analysis reveals a central contrast between high- and low-pressure prefectures. Following the 2017 tightening of cropland protection, urban expansion trajectories remain similar across prefectures, whereas cropland adjustment differs sharply by cropland protection pressure. Figure 2 presents prefecture-demeaned trends in impervious surface area and cropland area, separately for core and non-core areas and by cropland protection pressure.

Across both core and non-core areas, trends in impervious surface expansion are broadly comparable between high- and low-pressure prefectures, with steady growth and no visible

⁵ $\sum_{c,p} (X_{cp} - \bar{X})^2 = \sum_p n_p (\bar{X}_p - \bar{X})^2 + \sum_p \sum_{c \in p} (X_{cp} - \bar{X}_p)^2$.

divergence around 2017. This pattern indicates that urban development proceeded along similar paths regardless of cropland protection pressure, in line with the continued prioritization of economic growth objectives.

In contrast, cropland dynamics diverge markedly by cropland protection pressure after 2017. Prior to the policy tightening, cropland area declines in both high- and low-pressure prefectures. After 2017, however, cropland area rebounds and increases in high-pressure prefectures, while continuing to decline and exhibiting only a limited and delayed recovery in low-pressure prefectures. This divergence is concentrated in non-core areas; cropland trends within core urban areas remain similar across prefecture types throughout the period.

These patterns suggest that the 2017 policy tightening primarily constrained high-pressure prefectures, while having limited effects in low-pressure prefectures. At the same time, urban expansion in core areas appears insulated from these constraints, implying that the burden of meeting binding cropland protection targets was shifted toward non-core areas within high-pressure prefectures.

4.2 Vegetation Coverage over the Reclamation Cycle

Figure 3 illustrates the evolution of fractional vegetation coverage on newly reclaimed land as a function of years since reclamation. Pooling all newly reclaimed grids in the sample, mean FVC increases monotonically over time. This pattern indicates that vegetation coverage improves gradually following reclamation, indicating that FVC captures the intensity of agricultural use and land cultivation in the years after cropland creation.

Figure 4 further disaggregates these patterns by cropland protection pressure and policy period. For grids reclaimed prior to 2018, as well as for grids located in low-pressure prefectures, FVC exhibits a similar upward trajectory to that observed in the full sample, increasing with years since reclamation. In contrast, newly reclaimed land in high-pressure prefectures after the policy tightening displays a markedly different pattern. These grids exhibit substantially lower initial FVC and show little subsequent improvement over time, with mean vegetation coverage remaining persistently low.

In a nutshell, these descriptive patterns suggest that while newly reclaimed land typically becomes increasingly cultivated over time, this process is disrupted for cropland created after the policy tightening in high-pressure prefectures. The persistently low vegetation coverage observed for these grids is consistent with lower post-reclamation cultivation or temporary fallow, whereas similar patterns are not observed in low-pressure prefectures. These motivating facts point to heterogeneous responses to the tightening of cropland protection, concentrated in regions with higher assigned compensation burdens.

5 Empirical Analysis

The empirical analysis uses cross-county variation in cropland protection pressure to estimate the effects of the 2017 enforcement tightening. The reform provides a common time shock, while predetermined differences in counties' exposure to delegated prefecture-level compensation obligations generate differential treatment intensity.

The analysis has four parts. First, the difference-in-differences analysis estimates effects on three primary outcomes: cropland expansion, soil suitability, and cultivation intensity. It combines an event-study design for cropland share with cohort difference-in-differences designs for the characteristics and subsequent use of newly reclaimed land. Second, a spatial regression-discontinuity design compares changes across the higher- and lower-pressure sides of provincial borders. Third, the subsection on additional results and alternative explanations documents other spatial and cropping patterns and tests whether seasonality, temporary land preparation, or crop switching can account for the cultivation results. Finally, the robustness analysis varies the construction of cropland protection pressure, newly reclaimed land, and fallow status.

5.1 Difference-in-Differences

5.1.1 Event-Study DiD: Cropland Area

The 2017 policy tightening constitutes a common shock across all counties, while cross-county differences in cropland protection pressure capture each locality's differential exposure to the tightening. A standard difference-in-differences design exploits this variation to identify the policy's effects on cropland expansion.

I estimate the following two-way fixed effects event-study specification:

$$\text{Cropshare}_{ict} = \sum_{k \neq -1} \beta_k (\text{Pressure}_c \times \text{Event}_{k,t}) + \mu_i + \lambda_t + \varepsilon_{ict}, \quad (2)$$

where Cropshare_{ict} denotes the cropland share of grid cell i in county c in year t . Pressure_c measures county-level cropland protection pressure, and $\text{Event}_{k,t}$ is an indicator for event time k , indexed relative to the first post-policy year, 2018. The omitted category is $k = -1$, corresponding to the 2017 reference year. Grid fixed effects μ_i absorb time-invariant spatial heterogeneity, while year fixed effects λ_t control for common shocks affecting all locations. The estimation sample excludes core urban districts, and standard errors are clustered by county.

The coefficients β_k trace out how the change in cropland share responds over time to the policy tightening as a function of cropland protection pressure. Identification relies on the assumption that, absent the policy change, cropland trends would have evolved similarly across counties with different levels of cropland protection pressure. The pre-policy period of the event study allows this parallel-trends assumption to be assessed directly.

5.1.2 Cohort DiD: Cropland Quality

While the event-study design captures aggregate changes in cropland area, it does not directly answer how the intensive margin of newly created cropland, namely its suitability and sub-

sequent use, is affected. Before a grid cell meets the reclamation criterion, post-reclamation suitability and utilization are not defined, so no comparable pre-reclamation outcome can be observed. An event study based on outcome levels before and after reclamation is therefore inapplicable. I instead study post-reclamation outcomes in a cohort-based difference-in-differences design, where a cohort is the set of grid cells first classified as reclaimed in a given year.

The sample comprises non-core grid cells first classified as reclaimed between 2013 and 2021. Each grid cell i , located in county c , is characterized by a reclamation year t_0 . For time-varying outcomes, cells are observed one to four years after reclamation. The outcome variable $Y_{i,s}$ captures post-reclamation outcomes along two intensive-margin dimensions: the suitability of newly created cropland and its subsequent utilization. The baseline empirical model is:

$$Y_{i,s} = \alpha_c + \mu_{t_0} + \delta_s + \beta(\mathbf{1}\{t_0 \geq 2018\} \times \text{Pressure}_c) + \varepsilon_{i,s}, \quad (3)$$

where α_c are county fixed effects, μ_{t_0} are reclamation-cohort fixed effects, and δ_s are years-since-reclamation fixed effects, which enter only for time-varying outcomes and absorb common post-reclamation dynamics. The coefficient β multiplies the interaction of a post-2017 reclamation indicator $\mathbf{1}\{t_0 \geq 2018\}$ with county-level cropland protection pressure Pressure_c , capturing differential exposure to the policy tightening among land reclaimed after the reform. Additional specifications interact baseline county population, GDP, land area, and government revenue with a linear calendar-time term. Standard errors are clustered by county.

Identification relies on a parallel-trends assumption: absent the policy tightening, the outcomes of newly reclaimed land in high- and low-pressure counties would not have diverged across reclamation cohorts. Appendix Table C.3 probes this assumption within the pre-policy sample by assigning false first-treated cohorts in 2014, 2015, and 2016 and re-estimating the baseline specification; none of the placebo estimates is statistically distinguishable from zero.

5.1.3 DiD Results

Figure 5 first establishes the extensive-margin response. Beginning in 2018, counties facing greater cropland protection pressure respond more strongly to the policy tightening and experience larger increases in cropland share than counties facing less pressure. All five post-policy estimates are positive and statistically significant. Before 2017, the estimates show no detectable differential pre-trend. None of the pre-policy coefficients is statistically significant. The data therefore provide no evidence that the parallel-trends assumption is violated. The contrast between the pre- and post-policy estimates supports interpreting the post-policy divergence as an effect of the tightening.

The estimates in Figure 5 are small in percentage-point terms because cropland reclamation is a rare event at the grid-year level. In any given year, only a small fraction of Jiangsu's land is converted to cropland, so the policy-induced increase in reclamation is smaller still when expressed as a change in cropland share averaged across all grid cells. To assess its economic significance, Appendix Table C.4 standardizes cropland protection pressure to mean zero and

unit standard deviation and benchmarks the resulting policy effect against historical cropland expansion. In 2018, a one-standard-deviation increase in pressure raises cropland share by 0.16 percentage points, equivalent to approximately 13,700 hectares in the non-core sample. This increase is 84 percent of the sample’s average annual gross cropland expansion during 2013–2017, approximately 16,200 hectares per year.

Table 2 turns from how much cropland counties created to what kind of land they created and whether they subsequently used it. Across both the 1995 China Soil Map and HWSO v2 2023, greater exposure to the tightening predicts a lower probability that newly reclaimed grids are suitable for cultivation. The estimates remain similar when baseline county characteristics are allowed to generate differential time paths, and the same post-policy cohorts are also more likely to remain fallow. The two outcomes capture distinct margins: soil suitability reflects the underlying agricultural quality of the selected location, whereas fallow status captures realized use after reclamation. Their joint movement therefore indicates that the response was not limited to expanding cropland onto less suitable land. High-pressure counties also brought into the cropland inventory plots that were less likely to be actively cultivated. The agreement across two soil classifications and a separate utilization measure makes it unlikely that the result is driven by a particular land-quality dataset or outcome definition.

The difference in post-reclamation cultivation is persistent rather than a short establishment delay. Appendix Figure B.6 shows that the fallow gap opens in the first year after reclamation and remains visible through the fourth year. If the result merely reflected the time required to prepare a newly reclaimed plot for cultivation, the gap should narrow as the land matures. Instead, the continuing difference shows that the additional land brought into the cropland inventory in high-pressure counties remains more likely to be fallow. The result therefore documents a sustained reduction in cultivation intensity rather than a short establishment delay.

Fallow status is defined only after reclamation, so the same binary outcome cannot provide a conventional pre-trend test. Figure 6 instead uses county-year vegetation coverage among grids that are ever reclaimed during the sample period. Before 2017, FVC follows similar paths across counties with different levels of cropland protection pressure. After the tightening, it declines more in high-pressure counties. This break provides an outcome available on both sides of the reform and therefore offers complementary support for the parallel-trends assumption underlying the cohort design. It also corroborates the fallow result with a continuous measure of realized vegetation, reducing the concern that the utilization finding is an artifact of the threshold used to classify land as fallow.

County-level output data provide complementary evidence on realized agricultural production. I divide the sum of grain, oilseed, and cotton output by satellite-measured cropland area and estimate a county-year event study using the paper’s cropland protection pressure measure, with county and year fixed effects. Appendix Figure B.7 plots the year-specific coefficients relative to 2017. The pre-policy estimates are small and statistically insignificant, and a joint test cannot reject that the 2012–2016 coefficients equal zero ($p = 0.741$). The estimate is -0.012

log points in 2018 and is marginally significant at the 10 percent level ($p = 0.085$). From 2019 through 2022, the estimates range from -0.025 to -0.030 log points and are statistically significant at the 5 percent level in every year. Crop output per cropland hectare therefore begins to decline with greater policy exposure in 2018, with a larger and persistent gap from 2019 onward. Because this outcome combines county-reported production with satellite-measured cropland area, I interpret it as complementary to the grid-level evidence on vegetation and fallowing.

The results show two linked responses to the policy tightening. More exposed counties expanded cropland rapidly after 2017, but the newly created land was less agriculturally suitable and less intensively cultivated. Tighter enforcement therefore increased the amount of land recorded as cropland while newly reclaimed plots were more likely to remain fallow and exhibited lower vegetation coverage.

5.2 Spatial Regression Discontinuity

5.2.1 Design

The county-level difference-in-differences design identifies differential responses to the 2017 policy tightening from cross-county variation in land protection pressure. I complement that analysis with a spatial regression-discontinuity design that asks whether the same response appears among locations that are geographically close but fall under jurisdictions with different protection pressure. The design compares changes in cropland outcomes among grid cells located on opposite sides of a provincial administrative border.

Appendix Figure B.8 depicts the difference in provincial cropland protection pressure across the border between Jiangsu and Anhui. Jiangsu faces greater provincial cropland protection pressure than neighboring Anhui. The main spatial RDD comparison is therefore between nearby grid cells on the Jiangsu and Anhui sides of their shared provincial border.

The analysis compares changes in cropland outcomes across adjacent counties along the border corridor between Jiangsu and Anhui. I construct pairs of counties that share the provincial border and form the analysis pool from non-mountain grid cells located within a 50 km corridor on either side of the border. The running variable is the signed distance from each grid-cell centroid to the provincial border, with positive values on the Jiangsu side and the cutoff at zero. The outcome, ΔY_i , is the cell's mean post-policy outcome minus its mean pre-policy outcome. I estimate the local linear specification

$$\Delta Y_i = \alpha + \tau J_i + \beta_1 d_i + \beta_2 (J_i \times d_i) + \mathbf{P}_i' \gamma + \varepsilon_i, \quad |d_i| \leq h, \quad (4)$$

where J_i equals one for a cell on the Jiangsu side, d_i is signed distance to the border, and $\mathbf{P}_i$ is a vector of cross-border county-pair indicators. The coefficient τ is the local Jiangsu-minus-Anhui discontinuity in the outcome change at the border. For each kernel, the county-pair indicators enter the local regression additively, and the MSE-optimal bandwidth is selected for that same covariate-adjusted estimator (Calonico et al., 2014, 2019; Cattaneo et al., 2024).

A central feature of the design is that the discontinuity is estimated in changes rather than in levels. For each grid cell, I define the dependent variable as the post-policy outcome relative to its pre-policy benchmark and estimate whether this change is discontinuous at the provincial border. The resulting estimator is thus a spatial RDD in outcome changes and allows cropland outcomes to differ in levels across the two provinces.

The identifying assumption is that, absent the 2017 tightening, expected changes in cropland outcomes would have evolved smoothly at the border. The design accordingly assumes parallel counterfactual changes among nearby grid cells on opposite sides of the border. Cell locations and the provincial boundary are fixed, so they cannot be chosen in response to the 2017 tightening. The main identification concern is that another contemporaneous factor may also have changed discontinuously at the border.

I probe the parallel-change assumption by estimating the same local linear RD separately for changes relative to 2012 in each year from 2013 through 2016. Appendix Table C.6 reports that none of the pre-policy cropland-share or fallow-rate discontinuities is statistically different from zero across the three kernels. The falsification evidence therefore supports the parallel-change assumption. Appendix Table C.7 further reports falsification tests for discontinuities in outcomes that the policy should not affect. Neither the pre-policy outcomes nor the policy invariant outcomes exhibit a statistically significant discontinuity at the border in the full sample of eligible grid cells.

5.2.2 Results

Figure 7 plots cropland-share changes at the Jiangsu and Anhui border. In each panel, the outcome is the cropland share in the indicated year minus the same cell’s average cropland share from 2012 to 2016.⁶ The discontinuity is small in 2017 and widens in later years. By 2022, the change in cropland share exhibits a positive discontinuity at the border, with a larger increase on the Jiangsu side than on the Anhui side. This spatial RD result is consistent with the DiD estimates: after the 2017 policy tightening, cropland expands more where cropland protection pressure is greater.

Figure 8 plots the change in fallow rate among cells that could potentially be used for cropland reclamation. The increase in fallowing is larger on the Jiangsu side, producing a positive discontinuity at the border. Thus, the higher pressure side exhibits both greater cropland expansion and a larger increase in fallowing among land that could be used to meet cropland protection requirements. Together, these findings show that higher-pressure locations record more cropland without a corresponding increase in sustained cultivation.

Table 3 reports the formal RD estimates. Across all three kernels, the Jiangsu-minus-Anhui discontinuities in cropland-share and fallow-rate changes are positive and statistically significant. Under the triangular kernel, the estimated discontinuities are 0.006 for cropland

⁶The figure ends the benchmark in 2016 so that the 2017 panel shows whether cropland-share changes were comparable across the border before full implementation. The formal RD estimates use 2012–2017 as the pre-policy benchmark, consistent with the main analysis, which treats 2017 as the reference year.

share and 0.037 for the fallow rate. These estimates confirm the patterns in Figures 7 and 8 and align with the DiD results: both designs show greater cropland expansion and increased fallowing where cropland protection pressure is higher.

As a robustness assessment, I examine whether the direction of the discontinuity follows the pressure ranking at the Jiangsu and Shanghai border. As shown in Appendix Figure B.8, Shanghai faces greater pressure than Jiangsu. Because the coefficients are defined as Jiangsu minus the comparison side, this reversed pressure ranking implies negative coefficients. Appendix Table C.8 reports the estimates for both outcomes across all three kernels. The coefficients are negative in every specification, as predicted by the reversed pressure ranking. The spatial RD and DiD estimates imply that higher-pressure locations expand cropland more after the policy but also experience greater fallowing among ex ante marginal cells.

5.3 Additional Results and Alternative Explanations

5.3.1 Additional Characteristics of Newly Reclaimed Land

Spatial Allocation and Fragmentation. Soil suitability and vegetation coverage may miss costs created by the location of new fields. If policy pressure pushes expansion toward the agricultural frontier, newly created cropland should be farther from existing fields and more isolated. I assess fragmentation at the expansion margin by examining where new cropland is placed relative to existing cultivation. Plots located near and connected to existing cultivation support more contiguous operations, reduce machinery movement between separated parcels, and facilitate mechanized, large-scale farming. China’s high-standard farmland criteria likewise emphasize land leveling, parcel consolidation, and field-road access as conditions for mechanized and large-scale production.⁷

To estimate whether the policy shifted where counties place newly created cropland, I measure the amount of newly created cropland in each 500-meter grid as the positive annual change in its pixel-derived cropland share.⁸ I then characterize the location of newly created cropland relative to the prior-year cropland landscape. The first outcome is the area-weighted mean distance from newly created cropland to the nearest grid with at least 1 percent cropland in the prior year. The second and third outcomes are the shares of newly created cropland located on grids with no cropland in the prior year and on isolated grids, respectively. I define an isolated grid as one for which the focal grid and all rook neighbors within the same county had cropland shares below 1 percent in the prior year. I construct all three outcomes at the county-year level and estimate a parsimonious $\text{Pressure}_c \times \text{Post}_t$ specification with county and year fixed effects and county-clustered standard errors, where Post_t equals one for annual transitions ending in each year from 2018 through 2022.

⁷See Appendix Section D.2 for the national high-standard farmland construction standard and the Ministry of Agriculture and Rural Affairs’ upgrading guidance.

⁸This measure differs from the main analysis, which identifies newly reclaimed land when a grid’s cropland share rises persistently from below 0.60 to above 0.70. The raw-change measure retains smaller within-grid changes omitted by that threshold definition. When constructing the outcomes, I weight each grid by the amount of newly created cropland.

Table 4 shows that stronger policy pressure shifts newly created cropland toward locations farther from and less connected to existing cultivation after 2017. Column (1) shows that, for a one standard deviation increase in pressure, newly created cropland is located 4.7 meters farther from the nearest grid with existing cropland ($p = 0.021$, BH-adjusted $q = 0.035$). Columns (2) and (3) show, respectively, that the share of newly created cropland located on grids with no cropland in the prior year rises by 0.52 percentage points ($p = 0.013$, BH-adjusted $q = 0.032$) and the share located on isolated grids rises by 0.29 percentage points ($p = 0.036$, BH-adjusted $q = 0.045$). These placement shifts indicate that stronger policy pressure redirects newly created cropland beyond the existing cultivated footprint and toward less connected locations.

Environmental Outcomes. Cropland compensation may impose environmental costs in addition to creating less productive or more fragmented farmland. Because expansion converts land previously classified as noncropland, it can replace water bodies, forests, and wetlands that provide ecological functions. I therefore examine whether stronger policy pressure increases the conversion of these land types into cropland. The annual CLCD data used to construct the paper’s cropland measure also identify each pixel’s land cover before conversion. I define a strict annual conversion when a 30 meter pixel is classified as water or forest in year $t - 1$ and as cropland in year t . I use GAWD to measure wetland to cropland conversion (see Appendix Table C.1 for the data source). Because the wetland measure may overlap the CLCD water and forest categories, the three measures should be interpreted as distinct outcomes based on different classification systems rather than as additive components of total land conversion. I average grid level conversion shares within each county and year and estimate the same parsimonious specification as above. The environmental sample covers annual transitions ending in 2013 through 2021 to maintain common data support across the three outcomes.

Columns (4), (5), and (6) of Table 4 show that the effects are concentrated in the conversion of water and wetlands. A one standard deviation increase in cropland protection pressure raises water to cropland conversion by 0.090 percentage points ($p < 0.001$, BH-adjusted $q < 0.001$) and wetland to cropland conversion by 0.042 percentage points ($p < 0.001$, BH-adjusted $q < 0.001$). The joint tests for coefficients before the policy do not reject for water ($p = 0.101$) or wetland ($p = 0.335$). In contrast, the estimated change in forest to cropland conversion is small and statistically insignificant at negative 0.011 percentage points ($p = 0.232$, BH-adjusted $q = 0.238$). These estimates indicate that cropland compensation imposes an environmental cost by redirecting expansion toward water and wetland areas.

5.3.2 Alternative Explanations

Three alternative explanations could weaken the interpretation of the main fallow result. First, low annual vegetation coverage may reflect normal seasonal variation rather than non-cultivation when crops should grow. Second, newly reclaimed plots may require a temporary preparation period before entering production. Third, they may be planted with crops that pro-

duce different vegetation profiles. Each explanation would weaken the conclusion that tighter cropland protection reduced post-reclamation cultivation. Appendix Sections A.1 and A.2 test these alternative explanations.

The first alternative is normal seasonality. To distinguish seasonal vegetation from fallow, I use pre-policy vegetation profiles and agronomic evidence to recover Jiangsu’s crop calendar, then ask whether post-policy fallow rises when fields are normally bare or during the growing season, when crop canopies should be dense. The latter pattern appears in the data. Relative to reclaimed land from pre-policy cohorts, post-policy reclaimed land has higher fallow rates in both high- and low-pressure counties, but the growing-season increase is consistently larger in high-pressure counties. This evidence supports weaker cultivation when crops should be growing, rather than unusually sparse vegetation during the lean season (Appendix Section A.1).

The second alternative is temporary land preparation. If newly reclaimed land requires time to enter production, the fallow difference should be largest immediately after reclamation and then decline. Cropping pattern maps allow me to follow newly reclaimed land through the first three years and distinguish detected cultivation from fallow.⁹ Cultivation and fallow do not differ clearly during the first two years, whereas lower detected cultivation and greater fallow emerge in the available third-year comparison. The delayed emergence of these gaps is inconsistent with a temporary land-preparation account (Appendix Section A.2).

The third alternative is crop switching. If a different crop mix accounts for lower vegetation coverage, the land should remain detectably cultivated while its cropping frequency or crop composition changes. Among cultivated land, I find no clear shift along either margin. Nor do the data indicate that crop switching accounts for lower vegetation coverage (Appendix Section A.2).

5.4 Robustness Checks

5.4.1 Sensitivity to Alternative Definitions

I first present a series of sensitivity checks using alternative definitions of cropland protection pressure, newly reclaimed land, and fallow status.

First, I examine whether the findings depend on how cropland protection pressure is measured. I consider alternative constructions of the pressure index by (i) redefining core-urban districts in the obligation component and (ii) modifying the definition of available land in the total-land component. These exercises test whether the results are driven by specific coding choices in the pressure measure.

Second, I evaluate the sensitivity of the results to the classification of newly reclaimed land and fallow status. I vary the threshold used to define reclamation based on pre-reclamation cropland share and allow for a range of alternative cutoffs. I also consider alternative definitions

⁹Fallow here refers to the fallow category in the cropping pattern maps, not the FVC-based definition used above. The maps partition newly reclaimed land into detected cultivation, fallow, and unclassified pixels. These categories are mutually exclusive and exhaustive.

of fallow land by varying the FVC threshold used to identify uncultivated plots. All robustness exercises are implemented for the main outcomes and are reported in the Appendix.

5.4.2 Robustness of the Cropland-Share Event Study

I then assess the robustness of the cropland-share event-study estimates to violations of parallel trends and county-specific linear time trends. I first apply the relative-magnitude restriction proposed by Rambachan and Roth (2023), which bounds changes in post-treatment deviations from parallel trends relative to the largest corresponding change observed before treatment. For the coefficient on a one-standard-deviation increase in cropland protection pressure, the break value of $\overline{M}$ lies between 0.75 and 0.80 (Appendix Figure B.9). The positive 2018 effect remains statistically significant when post-treatment changes in the deviation from parallel trends reach 75 percent of the largest corresponding pre-treatment change. The absence of detectable pre-trends and this bound make it unlikely that differential pre-policy trends explain the post-policy increase in cropland share.

Counties may follow different underlying time trends that could account for part of the estimated post-policy response. To address this concern, I re-estimate the event study with county-specific linear calendar-year trends, which absorb gradual differences in the evolution of cropland share across counties. This specification normalizes the 2016 and 2017 pressure-by-year interactions to zero: 2017 fixes the level, while 2016 anchors the linear slope. Appendix Figure B.10 shows that all five post-policy estimates remain positive and statistically significant at the 1 percent level. After rescaling pressure to one standard deviation, the estimated effect is 0.14 percentage points in 2018 and rises to 0.53 percentage points in 2022. County-specific linear trends therefore do not account for the post-policy increase in cropland share.

6 Mechanisms

Following the 2017 policy tightening, cropland expansion in high-pressure counties was accompanied by lower agronomic suitability and less intensive subsequent cultivation. The mechanism analysis addresses two questions. First, does the decline in reclamation quality reflect distorted reclamation choices? Costly compliance remains one possible explanation: local governments may expand sequentially along a land-quality frontier, so that the exhaustion of more suitable sites mechanically pushes additional reclamation toward marginal land. Distorted implementation, by contrast, arises when local governments depart from the best feasible reclamation path and select lower-quality land than resource constraints alone would require, potentially because identifying and developing better land requires greater effort or exposes implementation to closer quality monitoring. Section 6.1 presents several pieces of indirect evidence supporting the presence of distorted reclamation choices. Second, do the incentives facing local officials shape which parcels are reclaimed? Because China's performance-evaluation system requires officials to pursue multiple targets, reclamation choices may reflect both career concerns and

pressure from competing economic-growth objectives. Section 6.2 tests these two incentive channels.

6.1 Evidence on Distorted Reclamation Choices

6.1.1 Strategic Avoidance of Quality Monitoring?

If the allocation of reclamation sites is shaped by local bureaucratic incentives, including performance-evaluation concerns, or is otherwise distorted, the locations selected for reclamation should vary with spatial differences in land-quality monitoring intensity. I examine this possibility using distance to the nearest province-level cropland-quality monitoring station as a proxy for monitoring exposure. The analysis asks whether governments avoid parcels near monitoring stations, where land quality is more observable, and whether the 2017 tightening of cropland-protection enforcement mitigated such avoidance. These tests provide indirect evidence on whether quality monitoring shapes where land is reclaimed and whether tighter area enforcement weakens this avoidance pattern.

Cropland-quality monitoring stations are long-term fixed sites that produce standardized information on soil and crop conditions for provincial monitoring, spot checks, databases, and annual reports.¹⁰ Because monitoring is spatially local (see Appendix Section D.2), greater distance indicates weaker exposure to higher-level quality oversight. I restrict the analysis to stations operating before 2017, so their locations predate the reform.¹¹ Their siting follows explicit representativeness and density criteria: national rules require monitoring stations to represent local soil types and cropping systems, while Jiangsu imposes minimum coverage-density requirements.¹² Station placement therefore centers on local agro-ecological representativeness and coverage density rather than officials' short-run reclamation choices. To examine whether these siting decisions confound the estimated relationship between monitoring exposure and reclamation choices, I also report a coefficient-stability test that adds a broad set of covariates potentially related to station siting.

I test whether, before and after the policy tightening, distance to the nearest monitoring station predicts two patterns: (i) the extensive margin—grid cells farther from a station are more likely to be reclaimed; and (ii) the intensive margin—among newly reclaimed grid cells, soil suitability declines with distance. I estimate:

$$Y_{it} = \theta_E \log(d_i) + \beta_E [\log(d_i) \times \text{Post}_t] + X_i' \gamma_E + \alpha_{c(i)} + \lambda_t + \varepsilon_{it}, \quad (5a)$$

$$Y_i = \theta_I \log(d_i) + \beta_I [\log(d_i) \times \text{Post}_{t_0(i)}] + X_i' \gamma_I + \alpha_{c(i)} + \mu_{t_0(i)} + \varepsilon_i. \quad (5b)$$

Here d_i is the distance in kilometers from grid i to the nearest pre-2017 province-level monitoring

¹⁰See Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

¹¹Not all Jiangsu jurisdictions publicly report monitoring-station coordinates. The analysis therefore uses only grid cells in the reporting jurisdictions and excludes grid cells elsewhere. Appendix Figures B.11 and B.12 show, respectively, the sample's geographic coverage and the distribution of distances from every 100-meter grid cell in these jurisdictions to its nearest station.

¹²See Ministry of Agriculture of the People's Republic of China (2016) and Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

station. In equation (5a), Y_{it} indicates whether grid i is newly reclaimed in year t : θ_E measures the pre-policy relationship between reclamation probability and monitoring distance, while β_E measures how that relationship changes after the policy tightening. In equation (5b), Y_i is soil suitability for land first reclaimed in cohort $t_0(i)$: θ_I measures the pre-policy distance–quality gradient, while β_I measures how that gradient changes after the policy tightening.

Both equations control for baseline cropland and vegetation conditions, core-area status, county differences, and urban-center distance; equation (5a) includes year fixed effects, and equation (5b) reclamation-cohort fixed effects. Avoidance of monitoring implies $\theta_E > 0$, stronger post-tightening avoidance implies $\beta_E > 0$, and a negative distance–quality gradient implies $\theta_I < 0$; β_I captures its post-tightening change.

Table 5 reports the results. Even before the policy tightening, grid cells farther from monitoring stations were more likely to be reclaimed ($p < 0.001$). The policy tightening strengthens this pattern ($p < 0.001$). Columns (3)–(4) replace the level of log distance with grid fixed effects, leaving the interaction estimates unchanged. Among newly reclaimed grid cells, greater distance predicts lower soil suitability under both the 1995 and 2023 measures (both $p < 0.001$). The policy tightening does not alter this distance–quality gradient ($p = 0.439$ and $p = 0.708$, respectively). This persistence is consistent with continued limitations in quality monitoring for reclaimed land farther from stations after the tightening.

To assess whether post-policy reclamation outcomes increase or decrease with monitoring-station distance, Appendix Figure B.13 plots residualized outcomes against physical distance to the nearest station. The outcomes are residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county fixed effects. Appendix Figure B.13 shows that the residualized annual reclamation probability increases with distance, whereas the residualized suitable-soil share among newly reclaimed grid cells decreases. These patterns reinforce Table 5 by showing directly that both reclamation and the quality of selected land vary with monitoring-station distance.

A potential concern is that monitoring-station siting may be correlated with other spatial conditions that also affect reclamation choices. To address this concern, Figure 9 examines coefficient stability after adding combinations of ten covariate groups covering administrative remoteness (county-boundary and government-seat distances); soil and baseline land cover around the nearest station (soil suitability and cropland, impervious-surface, forest, and water shares within 5 kilometers); terrain (elevation and ruggedness); and local economic activity (nighttime lights). The coefficients retain their signs across the displayed combinations, and the fully adjusted estimates remain statistically significant. The monitoring-distance gradients are therefore robust to a broad set of factors potentially related to station siting.

6.1.2 Do Governments Favor Previously Cultivated Land?

A government can meet a binding cropland-replacement target at lower cost by favoring parcels that are easier to bring into production. If local officials respond along this margin, the 2017 enforcement tightening should change not only total reclamation but also the composition of

the land selected: jurisdictions facing greater land pressure should shift more strongly toward low-cost sites. This sorting response would allow governments to satisfy the monitored acreage target while limiting the effort required per unit of newly created cropland. It therefore provides a separate test of whether implementation tilts toward easier sites rather than expanding uniformly along the available land frontier.

Prior evidence supports using cultivation history as a proxy for reclamation cost. A United Nations report estimates that restoring previously cultivated but abandoned land costs about \$100 per dunum, compared with \$180–200 for never-cultivated marginal land. A USDA report likewise documents clearing costs of \$15–30 per acre for lightly vegetated, recently abandoned fields, compared with \$95–160 for woodland converted to crops (see Appendix Section D.4). Because parcel-level reclamation costs are not observed, I use prior cultivation to capture this cost margin. Using the same annual land-cover data as in the main analysis, I reconstruct each 500-by-500-meter grid’s land-use history over 2001–2011. I classify a grid as previously cultivated if it meets the paper’s cropland definition for at least two consecutive years during this period but is no longer classified as cropland in 2012. The comparison group consists of grids classified as noncropland throughout 2001–2012.

I test this prediction by asking whether reclamation shifts toward previously cultivated grids after 2017 and, if so, whether this shift is concentrated in high-pressure counties. The specification includes an interaction between previously cultivated status and the post-policy dummy, together with its further interaction with cropland protection pressure. The latter triple interaction captures whether any post-policy shift toward previously cultivated land occurs disproportionately in high-pressure counties.¹³ Table 6 reports results under two definitions of previously cultivated land. Both require a grid to be noncropland in 2012, but classify it as previously cultivated if it was cropland for at least two or three consecutive years, respectively, during the 2001–2011 pre-policy period.

The table’s bottom rows translate the estimates into 1-SD magnitudes. At sample-mean Pressure, the post-policy effects are 0.036 and 0.039 percentage points and are statistically insignificant ($p = 0.173$ and $p = 0.184$). A 1-SD increase in Pressure raises the relative post-policy reclamation probability of previously cultivated land by 0.051 and 0.055 percentage points ($p = 0.010$ and $p = 0.015$). Adding these gradients to the average effects yields total effects of 0.087 and 0.094 percentage points at 1 SD above mean Pressure (both $p < 0.05$). Under both historical definitions, higher-pressure counties therefore become more likely to reclaim previously cultivated land after 2017, consistent with greater selection of lower-cost parcels.

¹³Specifically, I estimate $Y_{ict} = \beta_1(H_i \times \text{Post}_t) + \beta_2(H_i \times \text{Post}_t \times \text{Pressure}_c) + \alpha_i + \lambda_{ct} + \varepsilon_{ict}$, where Y_{ict} is 100 times the reclamation indicator for grid i in county c and year t ; H_i indicates previously cultivated land; Post_t equals one after 2017; and α_i and λ_{ct} are grid and county-by-year fixed effects. Standard errors are clustered by county.

6.1.3 Do Local Governments Move Sequentially Down the Land-Quality Frontier?

A decline in reclaimed-land quality may reflect costly compliance rather than distorted implementation. If a county develops the best feasible sites first, two changes can lower the quality of its next reclamation cohort. First, suitable land may become a smaller share of the remaining reservoir. Second, a large reclamation cohort may reach further down the quality frontier within the same year. I therefore define the *frontier component* as the quality loss predicted by both the beginning-of-year reservoir composition and the current reclamation quantity. The *selection wedge* is the amount by which observed cohort quality falls below this quantity-adjusted historical reference.

I estimate the reference using county-by-cohort observations from 2002 through 2017. The first predictor is the share of the factual beginning-of-year remaining reservoir classified as suitable by the 1995 soil measure. The second is current reclamation intensity: land reclaimed in year t divided by the factual reservoir remaining at the beginning of that year. The regression includes county and reclamation-cohort fixed effects, uses reclaimed-area weights, and clusters standard errors by county. I then insert each county's observed 2018–2021 reservoir composition and current reclamation intensity into the pre-policy relationship.

The remaining suitable share retains the sign implied by movement down a quality frontier. A one-percentage-point reduction predicts a 3.57-percentage-point reduction in the suitable share of the next cohort (standard error 1.42; $p = 0.015$). Current reclamation intensity does not have the predicted negative coefficient. Its estimate is 0.67 (standard error 1.75; $p = 0.703$). Thus, the pre-policy data do not show that larger same-year reclamation cohorts mechanically have lower average quality after conditioning on the reservoir state, county fixed effects, and cohort fixed effects.

Panel B of Appendix Table C.11 presents the aggregate accounting. Newly reclaimed land in 2018–2021 is 9.84 percentage points less suitable than the county-specific historical benchmark. The quantity-adjusted frontier component is 2.49 percentage points, compared with 2.43 percentage points when the reference uses only remaining quality. Of the adjusted frontier component, 2.42 percentage points come from deterioration in the remaining suitable share and only 0.07 percentage points come from current reclamation quantity. The selection wedge remains 7.35 percentage points, or 74.7 percent of the aggregate quality loss.

Panel C focuses on the additional decline in higher-pressure counties. A one-standard-deviation increase in cropland-protection pressure predicts a 5.63-percentage-point larger post-2017 quality loss. The quantity-adjusted frontier accounts for 0.09 percentage points, while the selection wedge accounts for 5.54 percentage points. The corresponding shares are 1.6 and 98.4 percent. The quantity contribution to the pressure-induced frontier estimate is -0.23 percentage points because the historical quantity coefficient is weakly positive and the pressure-related quantity path does not predict a quality decline. This negative contribution should not be read as evidence that reclamation improves land quality; it shows that the historical data provide no support for the proposed same-year quantity channel.

These estimates remain a back-of-the-envelope decomposition. Adding current quantity

raises the rolling pre-policy root mean squared prediction error from 27.32 to 29.88 percentage points, compared with 27.10 for a county historical-mean forecast. Only 40.6 percent of post-policy reclaimed area lies within the county-specific pre-policy support of both predictors. I therefore do not treat the component shares as precise structural parameters. The narrower result is that accounting for the amount reclaimed in the current year does not explain the post-policy quality decline. The large residual remains indirect evidence consistent with distorted reclamation choices rather than a point estimate of welfare misallocation.

6.2 Incentives Facing Local Officials

Section 6.1 provides several pieces of indirect evidence consistent with distorted reclamation choices. The following analysis asks whether the incentive scheme facing local officials helps explain these patterns. Officials are evaluated on multiple objectives, including cropland area compliance and economic growth, while the quality of newly reclaimed land is monitored less closely. The hypothesis is that officials prioritize observable and rewarded outcomes at the expense of the weakly monitored quality margin when career incentives are stronger or multiple objectives are more difficult to satisfy jointly. I test this hypothesis by asking whether land misallocation is greater when promotion incentives are stronger, when a new party secretary takes office, and when economic growth pressure is higher.

6.2.1 Career Incentives

The career-incentive hypothesis predicts greater hidden quality distortions in higher-pressure counties when bureaucrats face stronger career incentives. Under China’s cadre evaluation system, a good performance record on assigned tasks is crucial for future advancement. The hypothesis implies that prefecture party secretaries with stronger promotion motives would place more weight on observable tasks such as economic growth and cropland protection. They would in turn press subordinate counties more strongly to create compensation cropland while preserving urban expansion. Higher-pressure counties bear more of this delegated obligation and are thus expected to respond more strongly after 2017. When acreage is closely monitored but quality is not, this differential response is expected to appear as a larger decline in reclaimed-land suitability.

I measure prefecture party secretaries’ career incentives in two ways. First, in China’s cadre promotion system, promotion prospects fall sharply after age 57 (Huang et al., 2024). Panel A therefore measures high promotion incentives with an indicator equal to one when the prefecture party secretary is not older than 57 and the year precedes the 2020 nationwide land inspection. Second, Panel B uses party-secretary turnover to capture a distinct career motive. According to research on political turnover and cadre incentives (Li and Zhou, 2005; Cao et al., 2026), newly appointed secretaries have stronger incentives to establish a good performance record early in their tenure. These officials are expected to place more weight on observable tasks and generate larger distortions along the hidden quality margin. The turnover indicator

is constructed from official appointment dates.

Table 7 tests the prediction separately for these two measures by interacting each measure with county-level cropland protection pressure and the post-2017 indicator.¹⁴ The triple-interaction coefficient asks whether the post-2017 suitability gap between higher- and lower-pressure counties within a prefecture widens when the party secretary’s incentives are stronger. All specifications include county and reclamation-cohort fixed effects, county-specific linear trends, and controls for the party secretary’s age, a male indicator, and Jiangsu birthplace. Because the incentive measures vary at the prefecture level and the number of prefectures is limited, inference uses a prefecture-level cluster bootstrap.

Across both panels and both soil datasets, the triple-interaction estimates are negative and statistically significant at the 5 percent level or better. Higher-pressure counties experience larger post-2017 suitability losses when the party secretary has stronger promotion incentives or has recently taken office. Appendix Table C.12 also shows that the pressure-related increase in fallowing is larger on lower-quality reclaimed land. This pattern suggests that higher fallowing rates may partly arise because lower-quality land selected at the reclamation stage is less likely to be cultivated afterward. These results support a career-incentive mechanism in which compliance shifts toward observable acreage and distortion toward the less monitored quality margin.

6.2.2 Economic Growth Pressure

As discussed in Section 2, prefecture-level bureaucrats face a multitask problem involving local economic growth fueled by urban expansion and cropland protection. Economic growth pressure may amplify both cropland expansion and the decline in reclaimed-land suitability. Urban expansion converts cropland to construction use, thereby increasing compensation obligations. Prefectures at greater risk of missing their growth targets may therefore have stronger incentives to preserve urban construction while meeting cropland requirements through reclamation. As pressure to meet competing observable tasks rises, officials may devote less effort to the less observable quality margin. This conflict therefore predicts larger cropland expansion but lower-quality reclaimed land.

I measure economic growth burden at the prefecture-year level as the shortfall between target growth and predicted annual growth based on first-quarter GDP performance. A prefecture-year is classified as facing high economic growth burden when this expected shortfall is positive and ranks in the top tercile among Jiangsu prefectures that year. The specification compares prefectures with different cropland protection pressure before and after 2017 and asks whether the post-policy land-use response to pressure is larger in high-burden prefecture-years.¹⁵ Each

¹⁴For newly reclaimed grid cell i in county c , prefecture p , and reclamation cohort t , both panels estimate $Y_{icpt} = \beta_1 P_c D_t + \beta_2 P_c D_t M_{pt} + X'_{cpt} \gamma + \alpha_c + \lambda_t + \kappa_c t + \varepsilon_{icpt}$, where P_c is county pressure, D_t is the post-2017 indicator, and M_{pt} is the promotion-incentive indicator in Panel A and party-secretary turnover in Panel B. X_{cpt} collects the relevant lower-order terms and party-secretary controls; α_c , λ_t , and $\kappa_c t$ denote county fixed effects, reclamation-cohort fixed effects, and county-specific linear trends. The coefficient of interest is β_2 .

¹⁵For outcome observation i in prefecture p and year or reclamation cohort t , I estimate $Y_{ipt} = \beta_1 (P_p \times D_t) +$

regression includes prefecture fixed effects, the relevant year or reclamation-cohort fixed effects, and party-secretary controls; the cropland-share specification also includes grid fixed effects.

Table 8 shows that, after the policy tightening, high-burden prefecture-years with greater cropland protection pressure experience more urban expansion. Both the number and area of newly supplied construction-land transactions increase, including transactions on land converted from cropland (all wild-bootstrap $p < 0.001$). Under the one-for-one balance policy, this urban expansion requires an equivalent increase in cropland area. Cropland therefore also expands more in the non-core rural areas of these prefectures ($p = 0.033$). Yet the newly reclaimed land used for this compensation is less likely to be suitable for cultivation ($p < 0.001$). These results suggest that local governments maintain urban expansion while meeting the resulting acreage obligation through lower-quality reclamation.

7 Economic Magnitudes and Incidence

7.1 Agricultural Economic Magnitudes

The estimated land-use distortions reduce agricultural value through increased fallowing and lower land quality. I quantify their economic magnitude for a representative hectare of newly reclaimed cropland under a one-standard-deviation increase in cropland protection pressure and then aggregate the implied loss across Jiangsu’s post-2017 reclamation.

The monetary value of one administrative hectare of newly reclaimed cropland follows from the agricultural production function.¹⁶ The resulting value expression is

$$v^{ha}(F, Q) = \kappa(1 - F)Q^\Omega. \quad (6)$$

Here F is the fallow share and Q is land quality. Intuitively, part of an administrative hectare remains fallow, for example because land-use rights are not leased or transferred to farmers or agricultural operators. This share generates no agricultural output. The remaining share $1 - F$ enters cultivation. For the cultivated portion, agricultural value rises with predetermined land quality Q , with elasticity Ω after farmers optimally adjust non-land inputs. The scalar κ

$\beta_2(P_p \times B_{pt}) + \beta_3(D_t \times B_{pt}) + \beta_4(P_p \times D_t \times B_{pt}) + \rho B_{pt} + X'_{pt}\gamma + \alpha_i + \mu_p + \lambda_t + \varepsilon_{ipt}$, where P_p is the raw prefecture-level cropland protection measure, D_t is the post-2017 indicator, and B_{pt} is the high-economic-growth-burden indicator. X_{pt} includes party-secretary controls. The prefecture and period fixed effects absorb the main effects of P_p and D_t . Here α_i is included only for the grid-year cropland-share outcome, while μ_p and λ_t denote prefecture and outcome-specific year or reclamation-cohort fixed effects. The coefficient of interest is β_4 .

¹⁶Following Chen et al. (2023), suppose physical output on cultivated land is $y(q, l, x) = a[(ql)^\theta x^{1-\theta}]^\gamma$, where q is land quality, l is land input, and x collects labor, capital, and intermediate inputs. A farmer chooses the endogenous input x to maximize $py(q, l, x) - c_x x$. The first-order condition gives $x^*(q, l) = [pa(1 - \theta)\gamma/c_x]^{1/[1-(1-\theta)\gamma]}(ql)^{\theta\gamma/[1-(1-\theta)\gamma]}$. Substituting x^* back into the production function yields $y^*(q, l) = A(ql)^\Omega$, where $\Omega = \theta\gamma/[1-(1-\theta)\gamma]$ and A collects constants. This derivation gives the monetary value of the cultivated portion of reclaimed land. Setting $l = 1$ for a cultivated hectare and absorbing monetary conversion into κ gives cultivated value κQ^Ω . An administrative hectare also includes the fallow share F , which contributes zero agricultural value; multiplying cultivated value by $1 - F$ gives equation (6).

captures common agricultural productivity, output prices, input costs, and the conversion from physical output to monetary value. Following Adamopoulos et al. (2024), the calibration sets $\Omega = 0.615$.¹⁷

To link agricultural value to pressure, we combine the value expression with the reduced-form relationships between pressure and the two agricultural margins. Let z denote standardized cropland protection pressure. The reduced-form analysis gives $F(z) = \bar{F} + \hat{\beta}_F z$ and $Q(z) = \bar{Q} + \hat{\beta}_Q z$, where $\hat{\beta}_F$ and $\hat{\beta}_Q$ are the corresponding estimates. Since $\bar{v}^{ha} = \kappa(1 - \bar{F})\bar{Q}^\Omega$, the value of an administrative hectare at pressure z can be written without separately observing κ :

$$v^{ha}(z) = \bar{v}^{ha} \frac{1 - F(z)}{1 - \bar{F}} \left(\frac{Q(z)}{\bar{Q}} \right)^\Omega.$$

Differentiating this expression gives the marginal agricultural cost of pressure:

$$MC^{ag} \equiv - \left. \frac{\partial v^{ha}(z)}{\partial z} \right|_{z=0} = \bar{v}^{ha} \left[\frac{\hat{\beta}_F}{1 - \bar{F}} - \Omega \frac{\hat{\beta}_Q}{\bar{Q}} \right]. \quad (7)$$

Table 9 reports the exact discrete loss from a one-standard-deviation increase in pressure:

$$Loss_{1sd}^{ag}/ha = \bar{v}^{ha} \left[1 - \frac{1 - (\bar{F} + \hat{\beta}_F)}{1 - \bar{F}} \left(\frac{\bar{Q} + \hat{\beta}_Q}{\bar{Q}} \right)^\Omega \right]. \quad (8)$$

The monetary baseline matters for the absolute magnitude of the loss. Baseline value is RMB 49,856 per hectare when measured using primary-industry value added, and RMB 31,156 per hectare when measured using agricultural value added. Across the reduced-form inputs reported in Table 9, the implied loss is 13.2–14.2 percent of baseline agricultural value. This is RMB 6,590–7,059 per hectare under the primary-industry value-added baseline and RMB 4,118–4,411 per hectare under the narrower agricultural value-added baseline. Appendix Table C.13 reports all numerical inputs.

Jiangsu aggregate loss. The per-hectare loss can be scaled using post-2017 newly reclaimed cropland and county-level excess pressure. Let

$$A_c = \sum_{t=2018}^{2022} 25 \Delta \text{CropShare}_{ct}$$

over grids crossing from crop share below 0.60 to above 0.70, and let $z_c^+ = \max\{z_c, 0\}$, where z_c is standardized cropland protection pressure. The aggregate loss is

$$Loss^{JS} = \sum_c A_c z_c^+ \times Loss_{1sd}^{ag}/ha.$$

¹⁷The calibration uses $\theta = 0.533$ and $\gamma = 0.75$, which imply $\Omega = \theta\gamma/[1 - (1 - \theta)\gamma] = 0.615$.

This gives $\sum_c A_c = 14,624$ hectares and $\sum_c A_c z_c^+ = 8,284$ pressure-weighted hectares after 2017. The implied Jiangsu loss is RMB 55–59 million using primary-industry value added, or RMB 34–37 million using agricultural value added, in 2017 prices.

7.2 Marketization and the Incidence of Cropland Protection

An important policy question is how to reduce the cropland quality and productivity losses induced by binding local land constraints. When enforcement tightens in land-scarce jurisdictions, meeting the compensation obligation through local reclamation pushes expansion toward less suitable plots. Quota trading offers an alternative: a jurisdiction can purchase supplementary-cropland indicators created elsewhere rather than meet the full obligation through local reclamation.

Jiangsu operates a provincial platform for such transfers (Department of Natural Resources of Jiangsu Province, 2023) (see Appendix Section D.3 for the earlier provincial rule). Yet this adjustment margin is costly. Local reclamation fees range from RMB 30 to 50 per square meter, whereas public platform results commonly report transaction prices of RMB 420 per square meter for paddy indicators and RMB 345 for non-paddy indicators (Jiangsu Provincial Real Estate Development Center, 2026). This gap raises a policy-design question: why does the trading system sustain prices far above direct reclamation costs instead of making quota purchases a more feasible and widely used compliance margin?

One hypothesis is that quota prices reflect the opportunity cost of reallocating cropland protection across space. Trading can spare a land-scarce buyer from reclaiming lower-quality plots, but the supplying jurisdiction must commit its own land to cropland protection. This commitment can restrict future urban expansion and the associated development opportunities. Consistent with this mechanism, Yu (2026) shows that cropland protection constraints restrict urban land supply and generate aggregate welfare losses. Lower transfer prices could therefore reduce quality losses in high-pressure buyers while concentrating cropland protection and its development costs in land-abundant suppliers. High prices may compensate suppliers for this spatial incidence, creating an equity-efficiency trade-off across jurisdictions.

To assess this opportunity-cost channel, I ask whether counties that absorb larger compensation obligations from core urban districts subsequently supply less construction land. Appendix Figure B.14 shows a relative decline in construction-land transaction supply after 2017 among higher-pressure counties. Appendix Table C.14 reports similar estimates for total transacted construction land, new construction-land supply, and possible cropland-origin new construction land. In the outer ring of the pre-policy built-up area, a one-standard-deviation increase in cropland protection pressure reduces these outcomes by approximately 12 to 16 percent after 2017. These estimates show that absorbing a larger cropland protection burden carries an urban-development cost. The same opportunity cost provides a rationale for high quota prices: suppliers must be compensated not only for creating replacement cropland but also for the development opportunities forgone when land is committed to cropland. High prices protect supplying jurisdictions from bearing this burden without compensation, but they

also limit quota trading’s ability to relax binding land constraints and reduce quality losses in high-pressure buyers.

8 Discussion and Conclusion

This paper examines how tighter enforcement of cropland protection shapes local land-use behavior in a centralized administrative system characterized by multitask incentives and imperfect monitoring. Leveraging cross-county differences in exposure to cropland protection pressure, the analysis shows that the post-2017 policy tightening generates responses along both the extensive and intensive margins of land use. Counties subject to higher pressure are more likely to expand cropland area, yet the land brought into cultivation is of lower soil suitability and is cultivated less intensively, with elevated fallow rates that persist over time. A spatial regression-discontinuity design, additional analyses, and tests of alternative explanations corroborate this pattern, while heterogeneity by career incentives, growth pressure, and monitoring exposure links it to local incentives and imperfect quality monitoring. These patterns suggest that stricter enforcement of observable quantity targets can induce distortions in land allocation and land use when monitoring of land quality and utilization remains limited.

The consequences of these distortions extend beyond short-run mismatches between land endowments and land-use categories. Reclamation farther down the local frontier is associated with inferior land characteristics, lower cultivation intensity, and higher fallow rates. Over time, these channels translate into economic losses that are not reflected in aggregate cropland area statistics. As a result, policies that succeed in stabilizing cropland quantity may nevertheless fail to achieve broader productivity objectives if they redirect land use toward plots with limited productive potential.

The findings also inform the design of policy implementation. A common response to execution distortions is to strengthen monitoring to additional dimensions, such as land quality and utilization, yet this approach may be constrained by administrative capacity. This study instead highlights the role of institutional arrangements, particularly the restricted functioning of cropland compensation quota markets. Because cropland compensation indicators do not circulate freely across prefectures, local governments facing binding targets have limited adjustment margins beyond local land reclamation. Greater flexibility and integration in cropland compensation indicator markets would enlarge the set of feasible compliance options, reducing the likelihood that multitask pressure results in distortive implementation. More broadly, the results underscore that mitigating misallocation under multitask governance requires not only tighter oversight, but also institutional designs that align incentives with policy objectives under realistic monitoring constraints.

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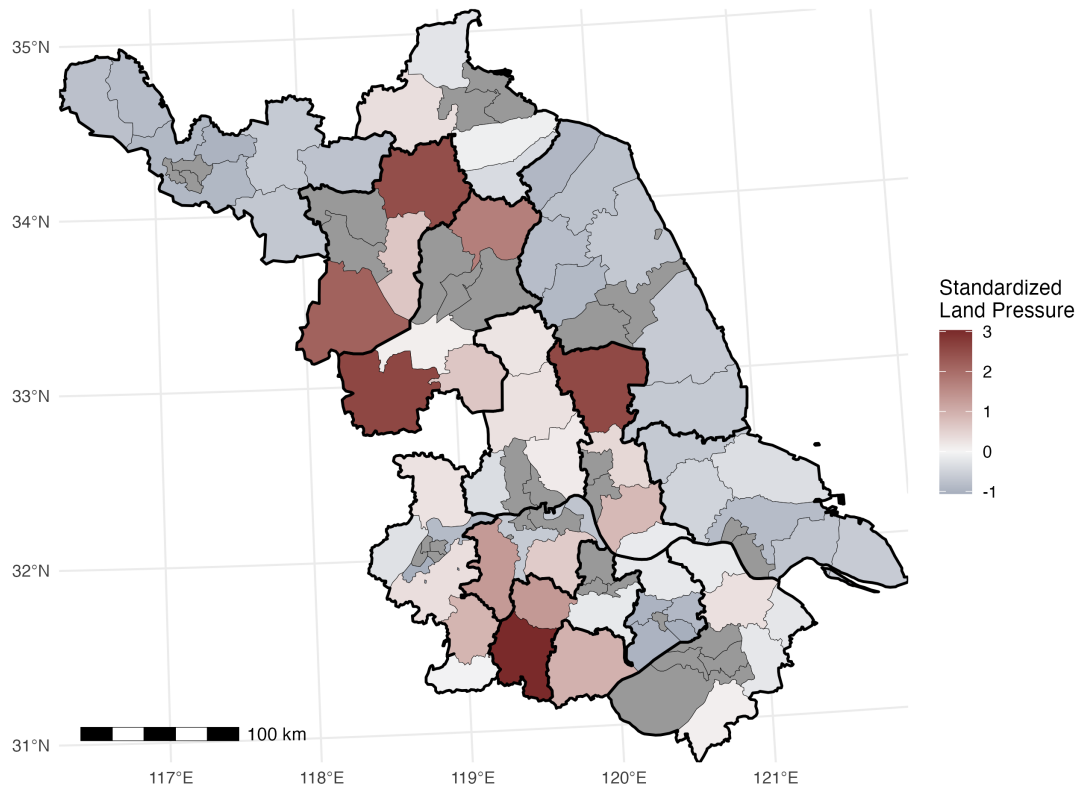
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Figures

Figure 1: Standardized Cropland Protection Pressure in Jiangsu



Notes: This figure maps the standardized county-level cropland protection pressure used in the heterogeneity analysis. The underlying measure is the ratio of the prefecture-level cropland reclamation obligation to the county's available non-core land. The plotted index is the z-score of this measure (mean zero, unit standard deviation) computed over the sample of non-core counties, so that positive values mark counties with above-average exposure to the cropland constraint and negative values mark counties with below-average exposure. Counties shaded grey are core urban districts and are excluded from the standardization sample. Because these districts primarily serve urban development and have limited scope for cropland reclamation, the compensation obligations they generate are unlikely to be fulfilled within the same districts. Thin black lines are county boundaries; thick black lines are prefecture boundaries.

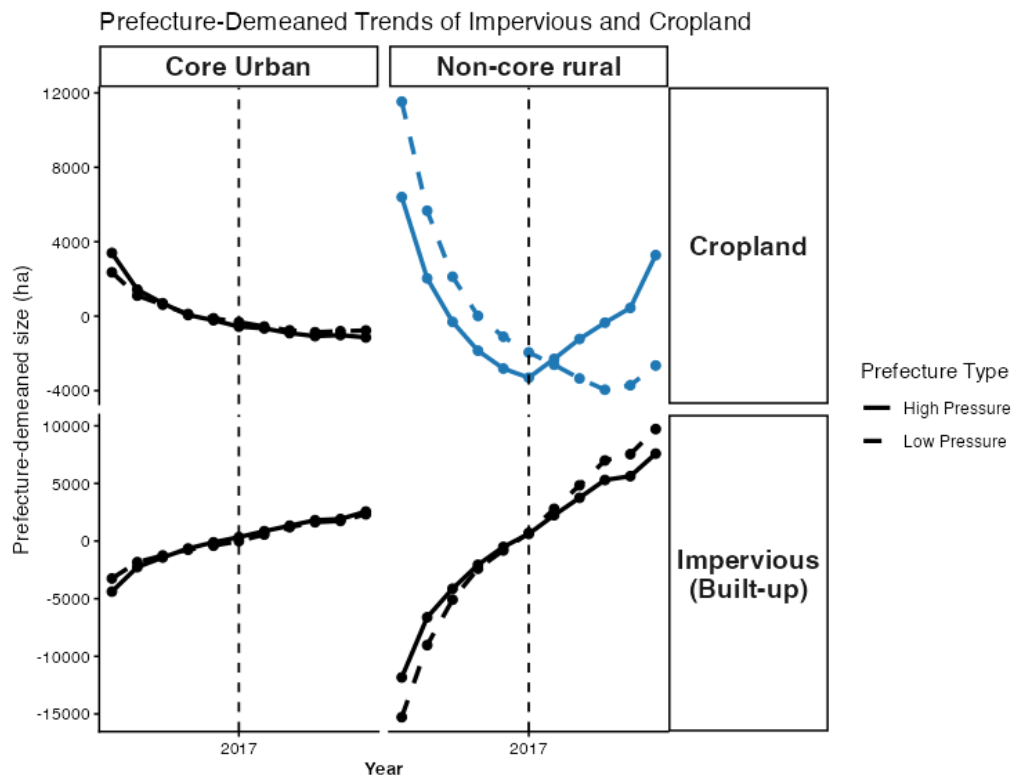


Figure 2: Trends in impervious area and cropland area by land type

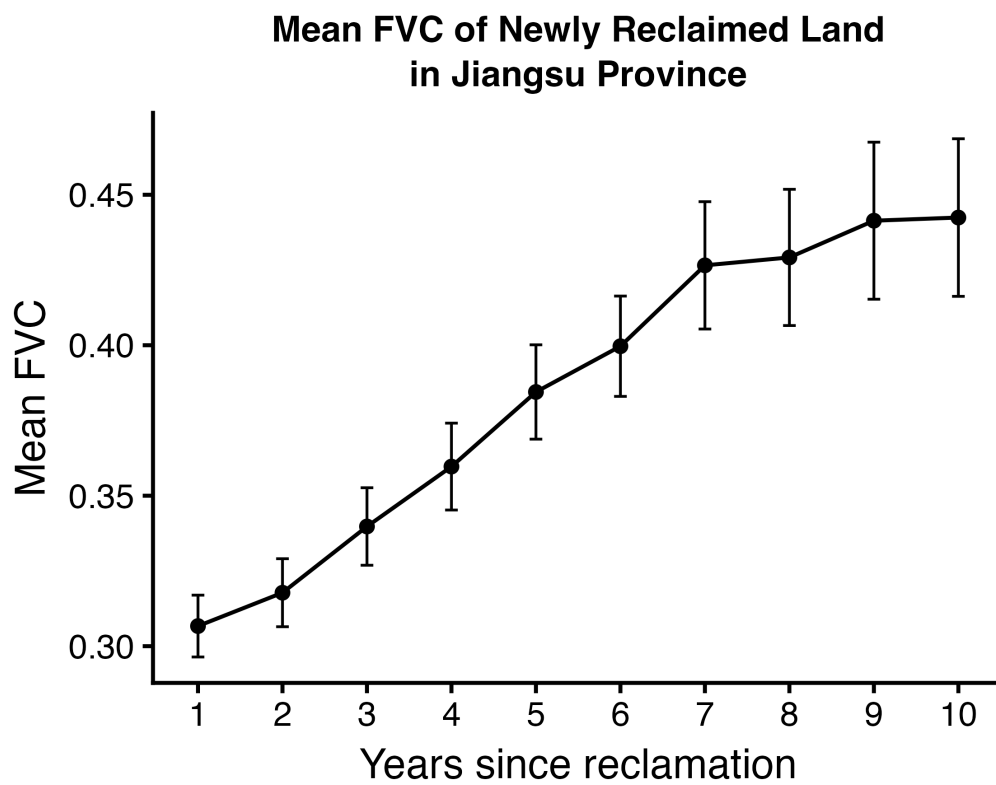


Figure 3: Evolution of Vegetation Coverage after Reclamation

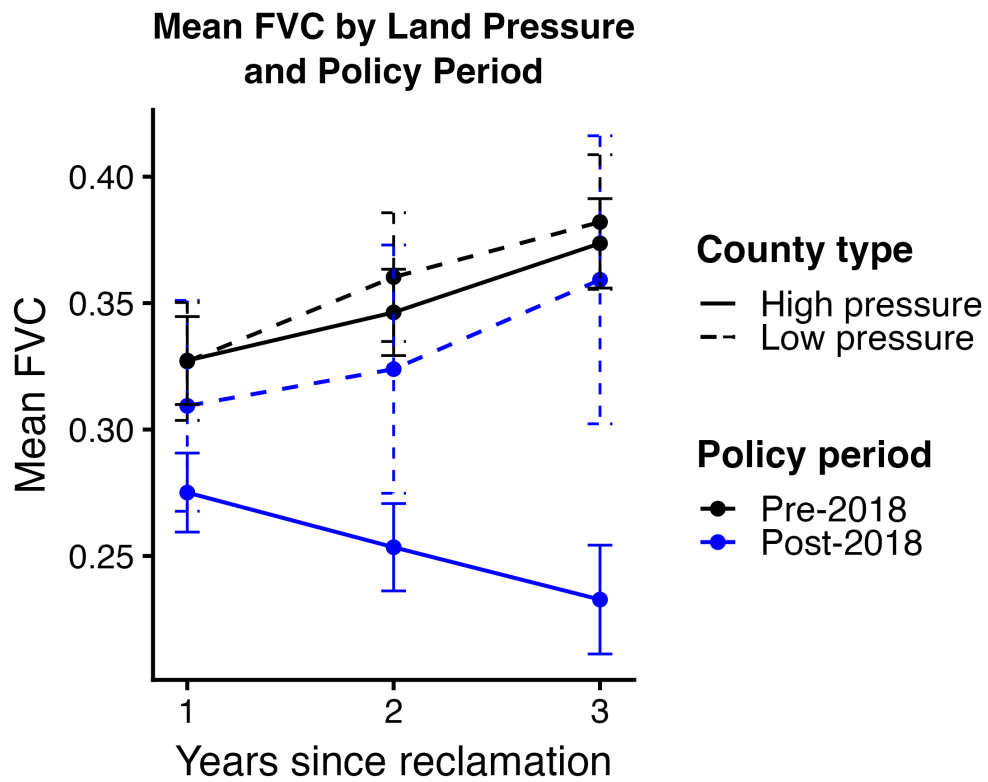


Figure 4: Vegetation Coverage after Reclamation by Cropland Protection Pressure and Policy Period

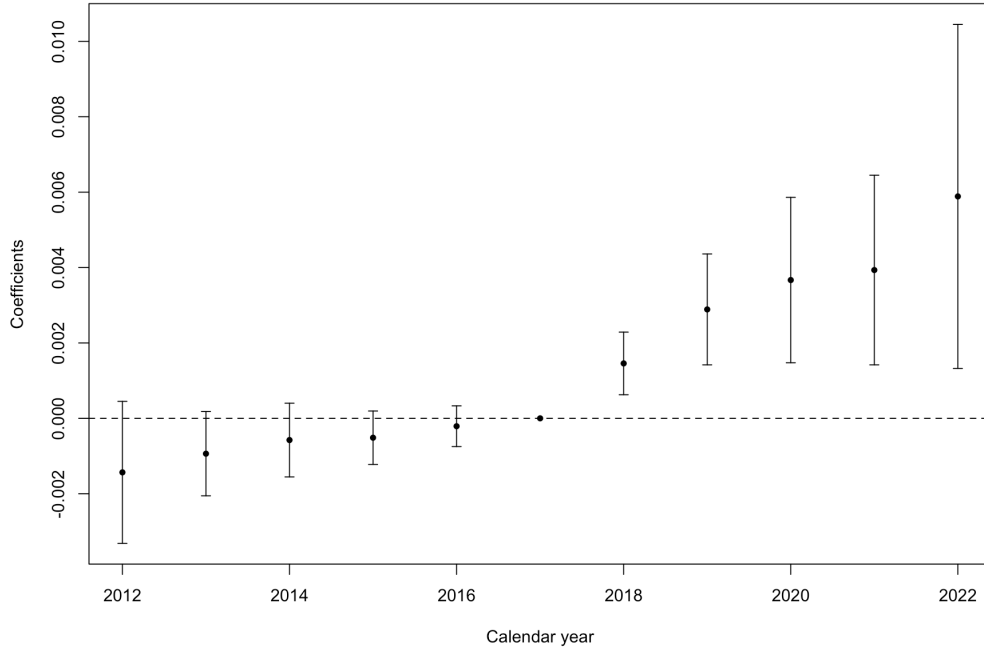
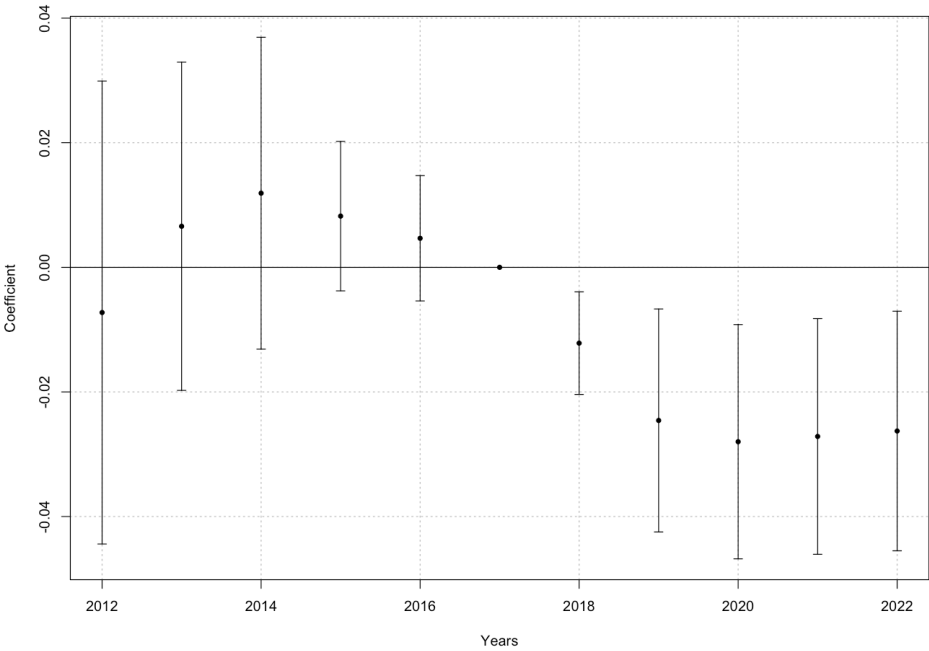


Figure 5: Effect on Cropland Share

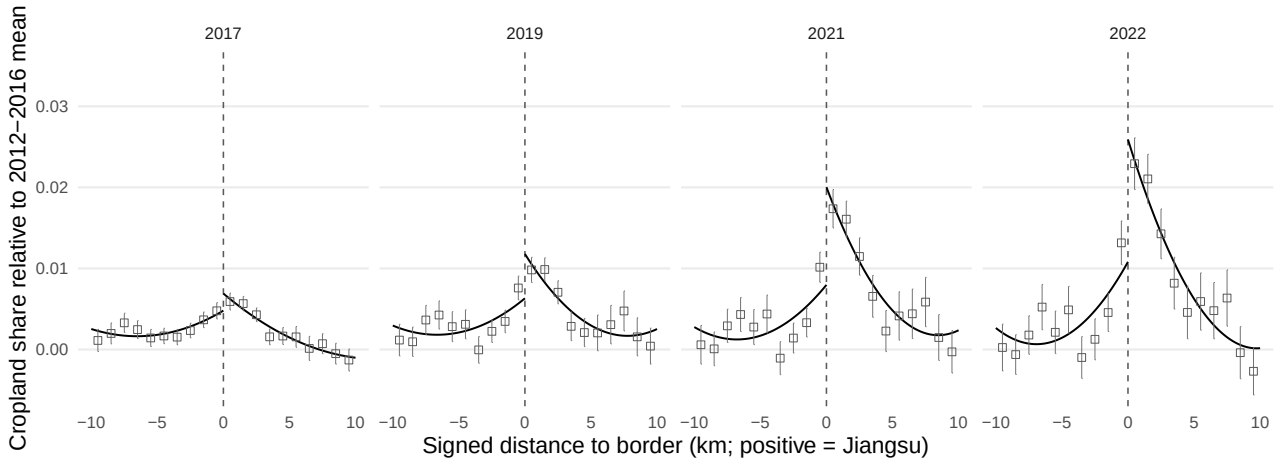
Notes: The figure plots coefficients on interactions between cropland protection pressure and calendar-year indicators in a panel of 500×500 meter grid cells in non-core counties. The outcome is cropland share; the specification includes grid-cell and year fixed effects, with 2017 omitted. Vertical bars show 95 percent confidence intervals based on standard errors clustered by county.

Figure 6: Effect of Cropland Protection Pressure on County-Level FVC among Ever-Reclaimed Grids



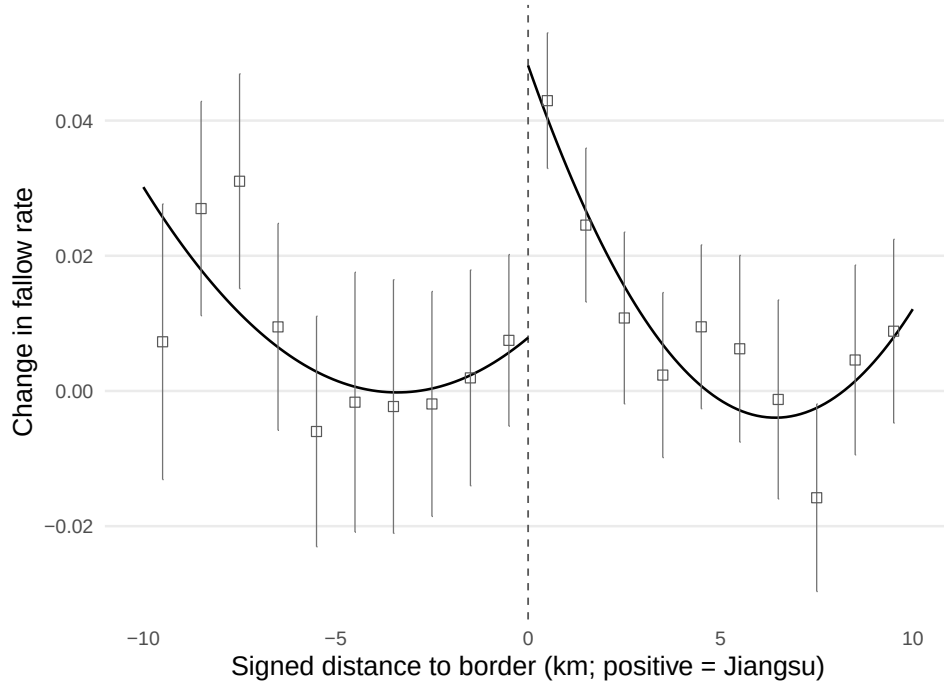
The figure plots coefficients from an event-study specification using county-year average FVC among grids that are ever newly reclaimed during the sample period. The county-year average is computed among grid-years with CLCD cropland share exceeding 0.6. Bars denote 95 percent confidence intervals.

Figure 7: Spatial RD in Cropland Share Changes at the Jiangsu and Anhui Border



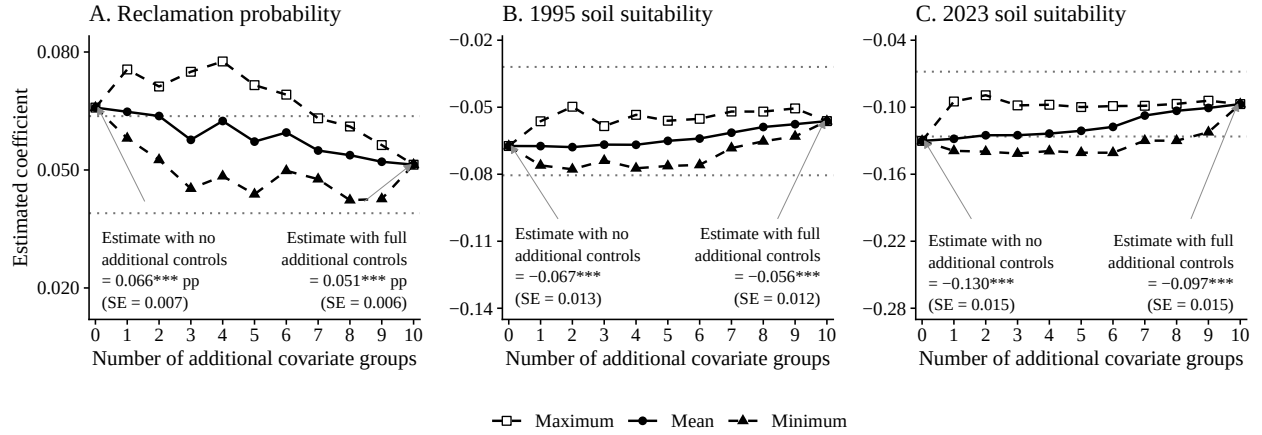
Notes: This figure plots spatial RD patterns in the change in cropland share at the Jiangsu and Anhui provincial border. The outcome in each panel is cell cropland share in year t minus the same cell's average from 2012 to 2016. The panels show 2017, 2019, 2021, and 2022. The sample comprises non-mountain cells in 13 cross-border county pairs within the 50 km border corridor. Because cropland share does not use FVC, the figure is not restricted by FVC availability. The plotted outcome is residualized by county-pair and year fixed effects. The running variable is signed distance to the border in kilometers, with positive values on the Jiangsu side. Points are 1 km bin means within a ± 10 km display window, and vertical bars report 90% confidence intervals. Solid lines are separate second order polynomial fits on each side of the cutoff for visualization; Table 3 reports the formal benchmark estimates.

Figure 8: Spatial RD in the Change in Fallow Rate at the Jiangsu and Anhui Border



Notes: This figure plots the spatial RD in the change in fallow rate at the Jiangsu and Anhui provincial border. The sample comprises non-mountain, ex ante marginal cells, defined by a 2012–2017 mean cropland share below 0.6, within cross-border county pairs. The outcome is the difference between the mean fallow rate from 2018 to 2022 and the mean fallow rate from 2012 to 2017. County-pair effects are removed within the ex ante marginal-cell sample for the binned visualization. The formal estimates include county-pair indicators directly in the local regression and do not condition on reclamation timing. The running variable is signed distance to the border in kilometers, with positive values on the Jiangsu side. Points are 1 km bin means within a ± 10 km display window, and vertical bars report 90% confidence intervals. Solid lines are separate second order polynomial fits on each side of the cutoff for visualization; Table 3 reports the formal benchmark estimates.

Figure 9: Sensitivity of Monitoring-Station Distance Estimates to Additional Covariates



Notes: Each panel holds the corresponding Table 5 specification fixed and adds subsets of ten covariate groups: county-boundary distance; soil suitability, cropland, impervious surface, forest, and water conditions within 5 kilometers of the nearest station; government-seat distance; elevation; terrain ruggedness; and nighttime lights. Panel (a) reports, in percentage points, the coefficient on $\log(d_i) \times \text{Post}$ for annual reclamation; panels (b) and (c) report the level coefficient on $\log(d_i)$ for the 1995 and 2023 soil suitability measures in outcome-share units. The zero-additional-group estimates reproduce Table 5 columns (2), (6), and (8). In panel (a), each added time-invariant covariate enters both in levels and interacted with Post; in panels (b) and (c), it enters in levels. For each number of added groups, the solid and dashed series report the mean, maximum, and minimum coefficient across the displayed specifications. The figure uses the unique specifications at zero and ten groups, all ten subsets at one and nine groups, and ten fixed random subsets at each count from two through eight; the same subsets are used across panels. Dotted lines show the parent-500-meter-clustered 95 percent confidence interval for the ten-group specification, and endpoint standard errors use the same clustering. The station-area soil measure is constructed from the 1995 soil map, and government-seat distance uses current verified locations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Tables

Table 1: Core Data Sources

Data	Source and coverage
Cropland compensation obligations and core-urban status	Prefecture Land Use Master Plans (2006–2020)
Land-cover data and county land shares	China Land Cover Dataset (CLCD), 30 m annual, 2012–2022; mainland China county boundaries, Harvard Dataverse
Fractional vegetation cover	National Tibetan Plateau Data Center, 30 m, 15-day, 2010–2022
Soil data	Soil Map of China, 1:1,000,000, National Soil Survey Office (1995); Harmonized World Soil Database v2.0, approximately 1 km (2023)
County socioeconomic data	China County Statistical Yearbooks

Table 2: Pressure, Land Suitability, and Fallow Status

	China Soil Map 1995		HWSD v2 2023		Fallow	
	1(Suitable)		1(Suitable)		1(Fallow)	
	(1)	(2)	(3)	(4)	(5)	(6)
Pressure $\times$ Post	-0.051** (0.020)	-0.057*** (0.021)	-0.033* (0.018)	-0.039** (0.019)	0.055*** (0.015)	0.055*** (0.017)
Control Mean	0.593	0.593	0.573	0.573	0.198	0.198
County controls $\times$ year		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE	—	—	—	—	✓	✓
Cluster level	County	County	County	County	County	County
Number of Counties	52	52	52	52	54	53
Num. Obs.	939	939	939	939	3,244	3,222
R ²	0.452	0.457	0.487	0.488	0.231	0.230

Notes: The table reports cohort difference-in-differences estimates from Model 3. Columns (1)–(4) observe each newly reclaimed grid once. In Columns (1)–(2), the suitable indicator equals one when the grid-level score from the 1995 China Soil Map is at least three. In Columns (3)–(4), the HWSD v2 2023 measure is the area-weighted mean of component-level suitable indicators, each equal to one when the corresponding suitability score is at least three. Columns (5)–(6) observe newly reclaimed grid cells by post-reclamation year and measure whether they are fallow. All specifications include county and reclamation-cohort fixed effects; the fallow specifications also include years-since-reclamation fixed effects. Even-numbered columns add baseline county population, GDP, government revenue, and land area interacted with year. Standard errors clustered by county are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Spatial RD Estimates at the Jiangsu and Anhui Border

	(1)	(2)	(3)
<i>Panel A: Δ Cropland share</i>			
RD in Δ cropland share (Jiangsu – Anhui)	0.006** (0.003)	0.006** (0.003)	0.007** (0.003)
<i>Panel B: Δ Fallow rate</i>			
RD in Δ fallow rate (Jiangsu – Anhui)	0.037*** (0.010)	0.038*** (0.011)	0.037*** (0.014)
County-pair indicators	Yes	Yes	Yes
Kernel	Triangle	Epanech.	Uniform

Notes. This table reports the formal local-linear RD estimates at the Jiangsu–Anhui border. The outcome in Panel A is the cell’s mean cropland share in 2018–2022 minus its mean in 2012–2017. The outcome in Panel B is the corresponding change in mean fallow rate. Panel A uses eligible non-mountain cropland cells in matched cross-border county pairs within the ± 50 km corridor. Panel B uses the subset with complete FVC and restricts the sample to ex ante marginal cells. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. County-pair indicators enter the local regression additively. Columns (1)–(3) use triangular, Epanechnikov, and uniform kernels, respectively; each column selects an MSE-optimal bandwidth. Entries are robust bias-corrected estimates. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table 4: Spatial Allocation and Environmental Outcomes of Newly Created Cropland

	Spatial allocation			Environmental outcomes		
	Dist. to existing cropland (m) (1)	Zero-cropland grids (2)	Isolated grids (3)	Water (4)	Forest (5)	Wetland (6)
Pressure $\times$ Post-2017	4.2** (1.8) [0.035]	0.47** (0.18) [0.032]	0.26** (0.12) [0.045]	0.082*** (0.018) [< 0.001]	-0.010 (0.008) [0.238]	0.038*** (0.009) [< 0.001]
<i>Magnitudes:</i>						
1-SD Pressure	4.7**	0.52**	0.29**	0.090***	-0.011	0.042***
Outcome mean	11.0	1.19	0.31	0.214	0.046	0.087
County fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Observations	640	640	640	576	576	576
R-squared	0.336	0.324	0.200	0.690	0.478	0.587
County clusters	64	64	64	64	64	64
Pre-policy joint p -value	0.317	0.600	0.288	0.101	0.366	0.335

Notes: All columns report county-year regressions for 64 counties. Columns (1)–(3) span 2013–2022; Columns (4)–(6) span 2013–2021 because the environmental measures end in 2021. Each regression includes county and year fixed effects; standard errors are clustered by county. Standard errors appear in parentheses, and BH-adjusted q -values appear in brackets. Post-2017 equals one from 2018 onward. Column (1) is the mean distance from newly created cropland to the nearest grid with at least 1% prior-year cropland, weighting each expanding grid by its newly created cropland area. Column (2) is the percentage-point share of newly created cropland on grids with zero prior-year cropland. Column (3) is the percentage-point share on isolated grids, defined as grids whose own and all within-county rook neighbors' prior-year cropland shares are below 1%. Columns (4)–(6) measure annual $t - 1$ to t source-to-cropland transition area as a percentage of observed valid CLCD area, averaged across grids within each county-year. The wetland measure uses GAWD and may overlap the CLCD water and forest measures. Outcome mean is the sample mean of the dependent variable. The pre-policy row reports the joint Wald-test p -value for the 2013–2016 pressure-by-year coefficients, with 2017 omitted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Cropland Quality Monitoring and Land Misallocation

	Extensive margin				Intensive margin: soil suitability			
	1(Reclaimed in year)				1995 China Soil Map		2023 HWSD v2	
	Level gradient		Grid FE					
	Baseline (1)	+ Urban (2)	Baseline (3)	+ Urban (4)	Baseline (5)	+ Urban (6)	Baseline (7)	+ Urban (8)
$\log(d_i) \times \text{Post}$	0.080*** (0.007)	0.066*** (0.007)	0.080*** (0.007)	0.066*** (0.007)	0.010 (0.014)	0.011 (0.014)	−0.004 (0.016)	−0.006 (0.016)
$\log(d_i)$	0.011** (0.004)	0.014*** (0.004)			−0.064*** (0.013)	−0.067*** (0.013)	−0.135*** (0.015)	−0.130*** (0.015)
Grid FE			✓	✓				
Year FE	✓	✓	✓	✓				
County FE	✓	✓			✓	✓	✓	✓
Reclamation-cohort FE					✓	✓	✓	✓
Baseline land-use controls	✓	✓			✓	✓	✓	✓
Urban-center distance control		✓		✓		✓		✓
Core grids included	✓	✓	✓	✓	✓	✓	✓	✓
Mean dependent variable	0.002	0.002	0.002	0.002	0.742	0.742	0.623	0.623
Observations	15,512,250	15,512,250	15,512,250	15,512,250	30,891	30,891	30,892	30,892
Parent 500m clusters	62,049	62,049	62,049	62,049	9,641	9,641	9,642	9,642
R^2	0.007	0.007	0.101	0.101	0.345	0.346	0.407	0.408

Notes: d_i is the distance in kilometers from 100-meter grid i to the nearest pre-2017 province-level cropland-quality monitoring station. Columns (1)–(4) use annual grid-year observations for 2013–2022, with Post = 1 in 2018–2022. Columns (1)–(2) estimate equation (5a) with county and year fixed effects and baseline land-use controls; columns (3)–(4) use grid and year fixed effects. Columns (5)–(8) use grids first reclaimed in 2013–2022 with county and reclamation-cohort fixed effects. Even-numbered columns add log urban-center distance, based on county centers constructed from 2012–2017 impervious-surface density: its level and Post interaction in column (2), its Post interaction in column (4), and its level in columns (6) and (8). Columns (1)–(4) report coefficients and standard errors in percentage points; columns (5)–(8) use outcome-share units. The 1995 and 2023 measures equal the share of soil-map components with a suitability score of at least 3. Baseline controls include 2012 cropland share and vegetation conditions. Standard errors are clustered at the parent 500-meter grid level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Selection of Previously Cultivated Land after the Policy Tightening

	$100 \times \mathbf{1}(\text{Reclaimed})$	
	(1) 2 years	(2) 3 years
Historical-cropland definition		
Previously cultivated land $\times$ Post	-0.022 (0.032)	-0.024 (0.035)
Previously cultivated land $\times$ Post $\times$ Pressure	0.046** (0.017)	0.050** (0.020)
<i>Magnitudes:</i>		
<u>Pressure</u>	0.036	0.039
1-SD Pressure	0.051**	0.055**
<u>Pressure + 1 SD</u>	0.087**	0.094**
Grid fixed effects	Yes	Yes
County-by-year fixed effects	Yes	Yes
Observations	385,390	374,218
R^2	0.222	0.220
County clusters	64	64
Estimation years	2013–2021	2013–2021
Pre-policy pressure-gradient p -value	0.241	0.301

Notes: The unit is an at-risk 500×500 -meter grid-year. The dependent variable is $100 \times \mathbf{1}(\text{Reclaimed})$, where Reclaimed equals one in the first year a grid is noncropland in $t - 1$ and cropland in both t and $t + 1$; coefficients are in percentage points. Previously cultivated land meets the main-text cropland definition for at least the indicated number of consecutive years during 2001–2011 and is classified as noncropland in 2012; comparison land is classified as noncropland throughout 2001–2012. The sample excludes fully urban-core county units and 2012 water-dominant grids. Joint models include the fixed effects listed in the table; standard errors are clustered by county. The corresponding event study additionally absorbs previous-cultivation-by-year fixed effects; its pre-policy p -value jointly tests the 2013–2016 pressure-gradient coefficients, with 2017 omitted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Multitask Incentives and Rural Cropland Suitability

	China Soil Map	HWSD v2
	1995	2023
	1(Suitable)	1(Suitable)
	(1)	(2)
<i>Panel A: Promotion-incentive mechanism</i>		
Pressure $\times$ Post	-0.097*** (0.020)	-0.096*** (0.027)
Party-secretary incentive $\times$ Pressure $\times$ Post	-0.058*** (0.012)	-0.056** (0.024)
Control Mean	0.593	0.491
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & Cohort FE	✓	✓
Number of Counties	52	52
Num. Obs.	939	939
R ²	0.498	0.453
<i>Panel B: Political-turnover mechanism</i>		
Pressure $\times$ Post	-0.094*** (0.031)	-0.094*** (0.022)
Party-secretary turnover $\times$ Pressure $\times$ Post	-0.045** (0.021)	-0.068*** (0.028)
Control Mean	0.593	0.491
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & Cohort FE	✓	✓
Number of Counties	52	52
Num. Obs.	939	939
R ²	0.498	0.454

Notes: This table reports mechanism tests relating political incentives to the quality of newly reclaimed cropland. The unit of observation is a newly reclaimed grid cell. Columns (1) and (2) use suitability indicators equal to one if the soil suitability score is at least 3, based on the 1995 China Soil Map and HWSD v2 2023, respectively. Both panels use the same heterogeneity specification: Panel A uses the party-secretary promotion-incentive indicator as the moderator, and Panel B uses the party-secretary turnover indicator constructed from official appointment dates. Party-secretary controls include age, a male indicator, and an indicator for Jiangsu birthplace. County linear trends are county-specific linear trends in reclamation cohort. Parentheses report exact Rademacher wild-cluster bootstrap standard errors clustered by prefecture; stars use the corresponding wild-bootstrap p -values. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Economic Growth Burden and Land Misallocation

	Rural cropland		Urban construction-land transfer value	
	(1) Cropland share	(2) 2023 suitable-soil indicator	(3) Log new- construction- source transfer value	(4) Log farmland-origin construction- source transfer value
Pressure $\times$ Post $\times$ economic burden	0.006 (0.003) [0.236]	-0.853*** (0.034) [0.004]	4.068** (0.638) [0.030]	4.263** (0.686) [0.027]
Grid FE	✓			
Prefecture FE	✓	✓	✓	✓
Year/cohort FE	✓	✓	✓	✓
Party-secretary controls	✓	✓	✓	✓
Observations	3,050,309	748	116	116
R^2	0.985	0.343	0.686	0.661
Clusters	13	13	13	13

Notes: This table reports DDD estimates of whether economic growth burden amplifies land misallocation. The coefficient of interest is Pressure $\times$ Post $\times$ Economic burden, where Pressure is the raw prefecture-level cropland protection measure and Post equals one after 2017. Economic burden equals one when a prefecture's predicted annual-growth shortfall is positive and ranks in the top tercile among Jiangsu prefectures in the same year. Columns (1)–(2) report rural cropland outcomes for newly reclaimed non-core cropland; column (2) uses the 2023 suitable-soil indicator. Columns (3)–(4) report $\log(1 + \text{prefecture-year aggregate transfer value in million yuan})$ for urban-core transactions contracted after March 31. Column (3) includes transactions whose recorded land source is new construction land. Column (4) further restricts these transactions to those whose centroid lies in a 500-meter grid containing cropland in the previous year; it is a proxy rather than a parcel-footprint measure. All specifications include the outcome-specific fixed effects listed in the table, all lower-order DDD terms, and party-secretary promotion-incentive and turnover controls. Parentheses report conventional standard errors clustered by prefecture. Square brackets report two-sided exact null-imposed, studentized Rademacher wild-cluster bootstrap p-values, enumerating all $2^{13} = 8,192$ sign patterns. Stars use these exact p-values. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Agricultural Economic Magnitude of Pressure

	No county controls	County trend controls
<i>Panel A. Reduced-form FVC-derived inputs</i>		
Fallow effect ($\hat{\beta}_F$)	0.059 (0.017)	0.060 (0.018)
Baseline fallow	0.198	0.198
<i>Panel B. Carried fixed non-FVC calibration</i>		
Suitability effect ($\hat{\beta}_Q$)	-0.049	-0.057
Baseline quality	0.491	0.491
Loss share (%)	13.2	14.2
Primary-industry loss (RMB/ha)	6,590	7,059
Agricultural VA loss (RMB/ha)	4,118	4,411

Online Appendix

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A Additional Results

A.1 Seasonality of Fallow Land Use

Fallow land remains uncultivated when crops would normally be grown. An annual FVC-based measure may overstate fallow if low FVC reflects sparse vegetation in non-growing months rather than non-cultivation during the growing season. I therefore examine monthly FVC and estimate month-specific effects to test whether the fallow associated with cropland protection pressure appears when crops should be growing.

To establish Jiangsu’s normal crop calendar before the policy intervention, I combine pre-policy vegetation data with agronomic evidence on the province’s planting seasons. The China Cropping Pattern maps confirm that multiple cropping is widespread and that double cropping is the dominant cropping pattern in Jiangsu Province.

Consistent with this double-cropping calendar, Panel A of Figure B.15 displays two high-canopy periods in the pre-policy FVC profile. FVC rises in April and May with the winter wheat canopy, falls in June as wheat is harvested and fields transition to rice, and rises again from July through September with the rice canopy.¹⁸ The same two-peak pattern appears for all cropland and for the rice and wheat rotation subset. Together, the pre-policy FVC profile and the crop-calendar evidence indicate that April–May and July–September are the high canopy windows, when winter wheat and rice, respectively, produce the highest vegetation cover.

I then test whether newly reclaimed land in high-pressure counties is more likely to be fallow during these high canopy windows. I estimate month-specific effects using the cohort DID specification in Model 3, with county, reclamation-cohort, and years-since-reclamation fixed effects.

Panel B of Figure B.15 reports monthly fallow-rate profiles and cohort DID estimates. The upper two plots show mean fallow rates for post-policy and pre-policy reclaimed cohorts separately in low- and high-pressure counties. In both pressure groups, post-policy reclaimed land is more likely to be fallow than the pre-policy benchmark throughout the year. The labels report these increases during the main growing seasons. They range from 6.5 to 13.4 percentage points in low-pressure counties and from 12.6 to 18.8 percentage points in high-pressure counties. The increase is therefore larger in high-pressure counties in each of the five high-canopy months. This contrast is the descriptive counterpart of the policy effect of interest, suggesting that tighter cropland constraints amplify the post-policy increase in fallow when active cultivation should produce dense crop cover. The lower plot tests the same hypothesis using continuous pressure and the cohort DID specification in Model 3. The estimates are positive in all five high-canopy months and statistically significant in April, July, and September, whereas the lean-season estimates are small and insignificant and June is negative.

Together, these results show that the FVC-based fallow effect captures reduced cultivation

¹⁸In 2015, rice and wheat account for 81 percent of harvested area and co-occur in 78 percent of harvest-positive grids. Crop-pattern maps classify 86 percent of non-fallow area as double cropped in 2015–2017. Rice and wheat rotation is therefore the principal benchmark, but not the only double-cropping system.

during the main growing seasons rather than normal seasonal variation in vegetation cover.

A.2 Crop Patterns and Fallow Land Use

The seasonal evidence shows that fractional vegetation cover-based fallow increases when crop canopies should be dense. A remaining explanation is that different crops or cropping frequencies produce lower vegetation cover on cultivated land. I use annual cropping-pattern maps to distinguish this explanation from lower detected cultivation.

The China Cropping Pattern maps (hereafter ChinaCP) classify crop activity at a 500-meter resolution from 2015 through 2021. Each pixel-year receives one mutually exclusive crop-pattern code.¹⁹ Within each reclaimed grid, I use CLCD to identify pixels that become persistent cropland, assign them the ChinaCP category observed in each post-reclamation year, and aggregate these matches into grid-event shares. The shares partition newly converted pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified. Detected cultivation is further divided into single, double, and triple cropping. All shares use the number of persistent newly converted pixels in the grid-event as their denominator; hence the frequency components sum to detected cultivation and the full partition to one.

I estimate Model 3 separately for the first three years after reclamation. The specification interacts continuous county-level cropland protection pressure with the post-policy cohort indicator, absorbs county, reclamation-cohort, and years-since-reclamation fixed effects, and clusters standard errors by county.

Panel A of Figure B.16 suggests a delayed reduction in detected cultivation on newly reclaimed land in more pressured counties. In the first two post-reclamation years, I find no significant differences in detected cultivation or assignment to the ChinaCP fallow category across counties with different levels of policy exposure. By the third year, however, detected cultivation falls by 8.9 percentage points and the ChinaCP fallow share rises by 5.5 points. Because another observed crop would remain part of detected cultivation, these offsetting changes indicate that more land is classified as fallow rather than shifted to another crop category.

Panel A speaks to a temporary crop-establishment explanation. Under this explanation, reclaimed land is initially uncultivated during preparation, so the fallow difference should peak immediately after reclamation and then decline. The estimates show the opposite: no significant difference in the first two years, followed by lower detected cultivation and more ChinaCP fallow in the third. Thus, observed fallow is not simply a short-lived consequence of land preparation.

Panel B asks whether crop patterns change among pixels with detected cultivation. I split this category into single, double, and triple cropping. Conditional on detected cultivation, this decomposition asks whether stronger policy exposure changes the number of crop cycles rather than whether the land is cultivated at all. The estimates show no clear differences: only

¹⁹ChinaCP codes combine cropping frequency and crop identity where available. The mutually exclusive classes are fallow; single maize, rice, wheat, or other crops; other double cropping; rice-maize; wheat-maize; double rice; wheat-rice; and triple cropping. Pixels without a valid code are labeled ChinaCP unclassified in this analysis.

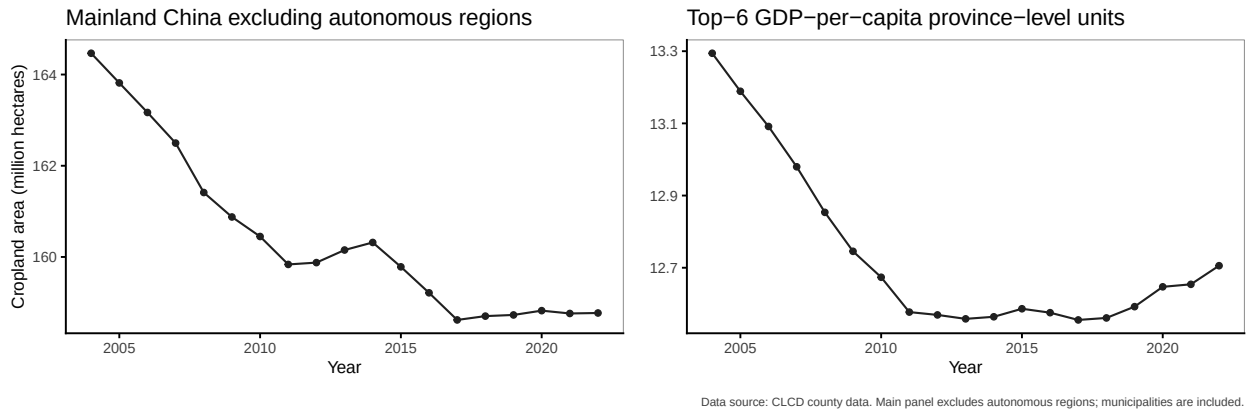
the first-year decline in double cropping is marginally significant. The remaining frequency estimates are statistically insignificant.

Panel C further decomposes the single- and double-cropping components into nine crop-specific ChinaCP classes. None of these coefficients is statistically significant at the 5 percent level in any of the first three post-reclamation years. Several estimates are marginally significant at the 10 percent level, but none remains significant after applying the Benjamini–Hochberg false discovery rate adjustment across the nine classes within each event year. The estimates therefore provide no evidence that stronger policy exposure shifts cultivation toward a particular observed crop or crop rotation.

The timing evidence in Panel A and the composition evidence in Panels B and C provide little support for either a short-lived establishment phase or a shift toward other observed crops or cropping frequencies. They are instead consistent with the main result: stronger policy exposure is associated with lower realized cultivation on newly reclaimed land.

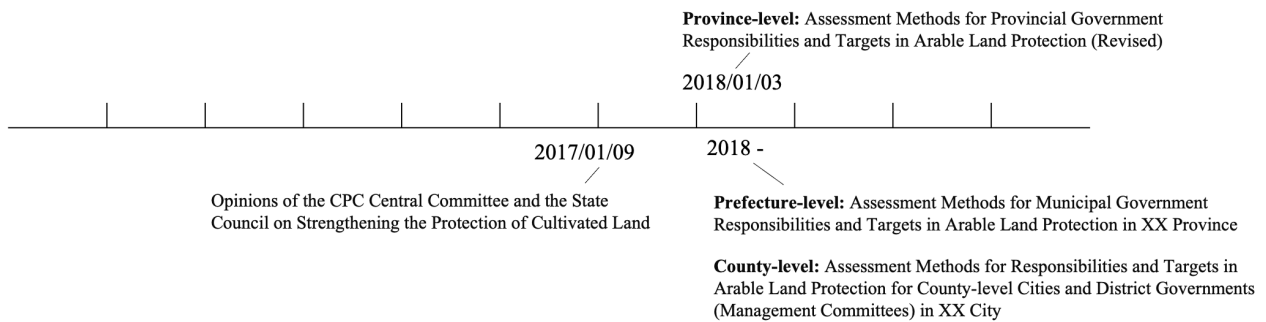
B Figures

Figure B.1: Satellite-Based Cropland-Area Trends in Mainland China



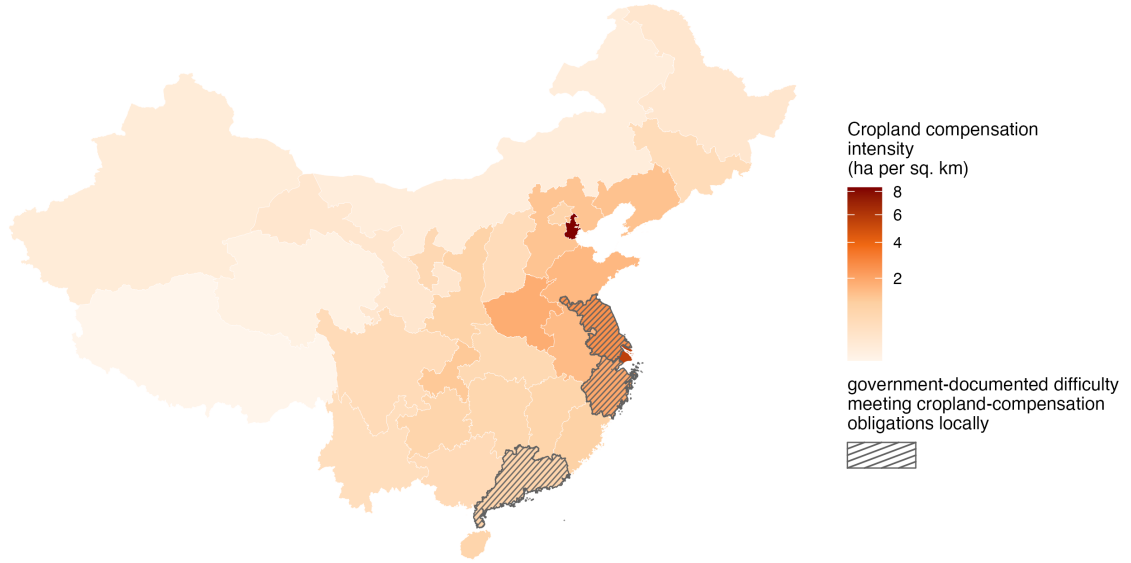
Notes: The figure reports cropland area aggregated from annual 30-meter China Land Cover Dataset classifications (Yang and Huang, 2021). CLCD is derived from satellite imagery. Panel A reports mainland China excluding the five autonomous regions (Inner Mongolia, Guangxi, Tibet, Ningxia, and Xinjiang); municipalities remain included. Panel B reports the six province-level units with the highest GDP per capita in 2021. Cropland area is measured in million hectares.

Figure B.2: Administrative Rollout of the 2017–2018 Policy Tightening



Notes: The 2017 Opinions strengthened cropland-protection accountability. Revised provincial assessment methods followed in January 2018, after which provincial and prefectural governments adopted corresponding measures for lower-level jurisdictions.

Figure B.3: Cropland Compensation Intensity



Notes: This figure maps the cropland compensation obligation assigned to each mainland province-level unit, divided by total provincial land area. The legend reports hectares of obligation per square kilometer of land area, equal to the source obligation measure in 10,000 hectares per square kilometer multiplied by 10,000. Darker shades indicate higher cropland compensation intensity, and gray areas indicate province-level units with missing obligation data. Jiangsu, Zhejiang, and Guangdong are identified as provinces where government documents report acute land-use constraints, scarce reserve land, or difficulty satisfying cropland-compensation obligations locally, as documented in Appendix Section D.3. Province boundaries are constructed by dissolving county-level polygons from the mainland China shapefile.

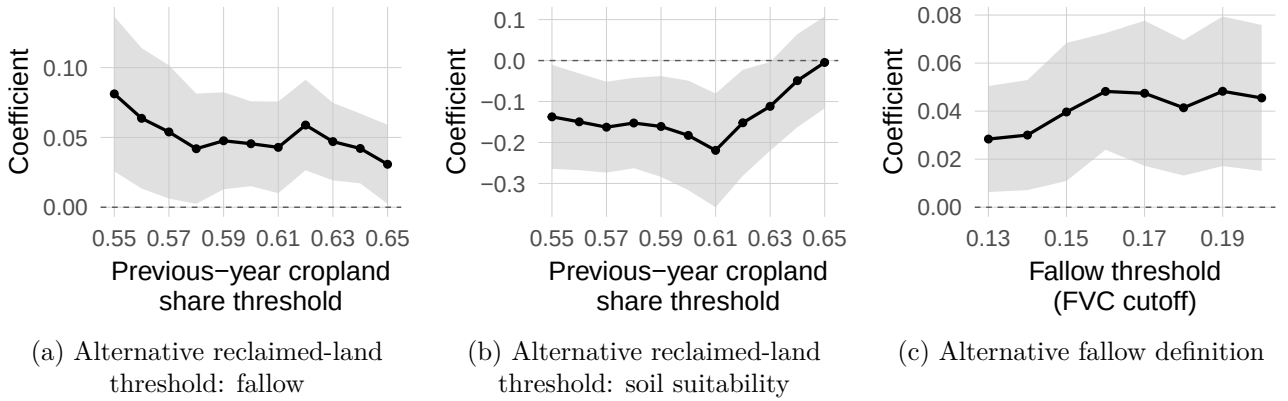


Figure B.4: Sensitivity to Alternative Definitions

Notes: Each panel plots the coefficient on $\text{Pressure} \times \text{Post}$ from the corresponding cohort difference-in-differences specification. Shaded bands report 95% confidence intervals. Panels (a) and (b) vary the previous-year cropland-share cutoff used to define newly reclaimed land. Panel (c) varies the FVC cutoff used to classify a newly reclaimed grid cell as fallow.

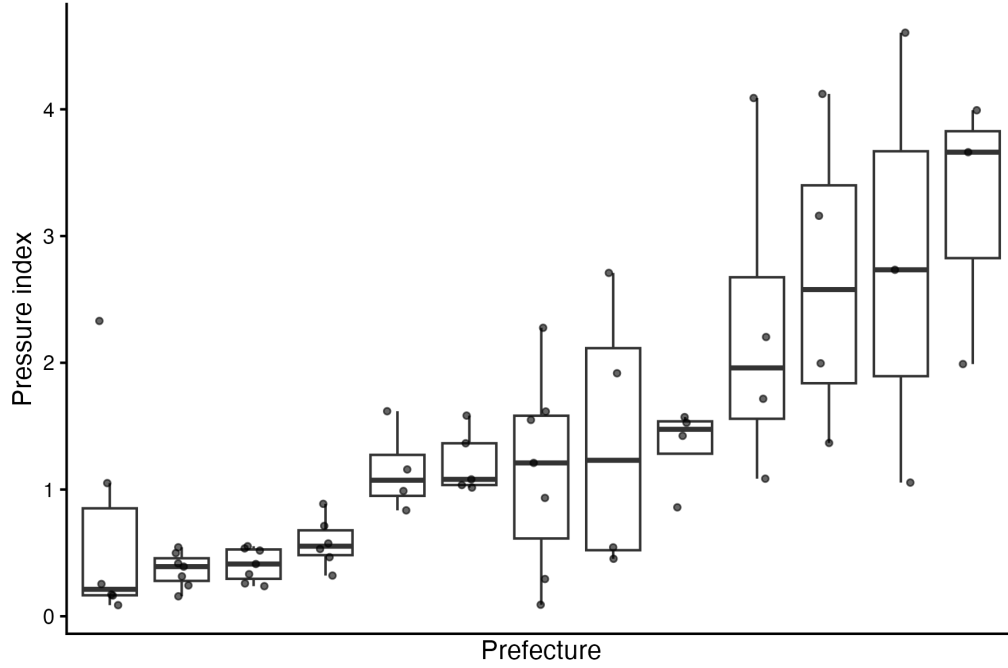
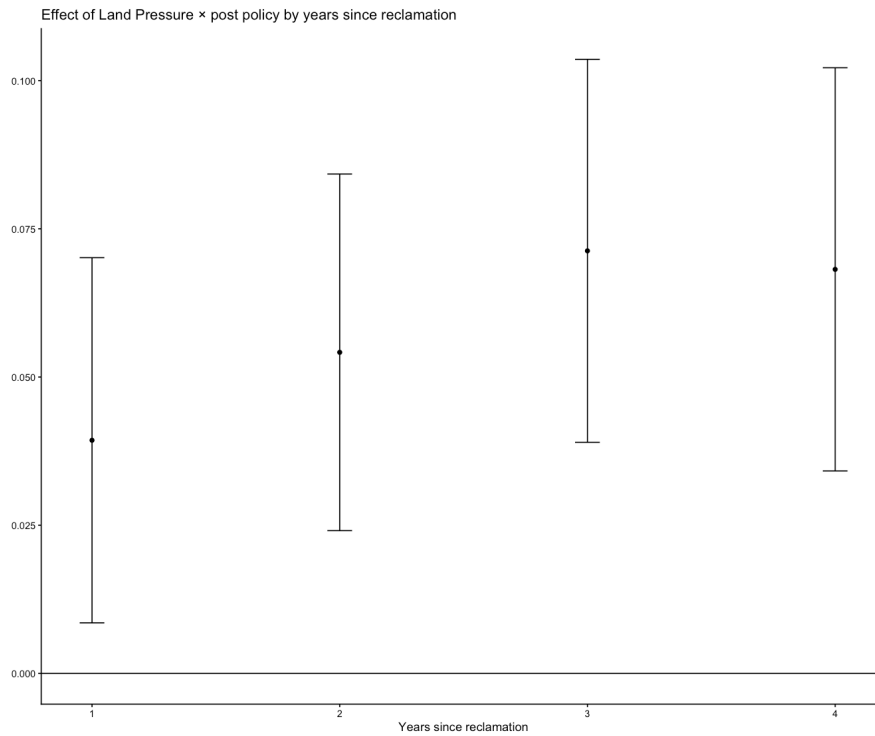


Figure B.5: Distribution of Cropland Protection Pressure within and across Prefectures

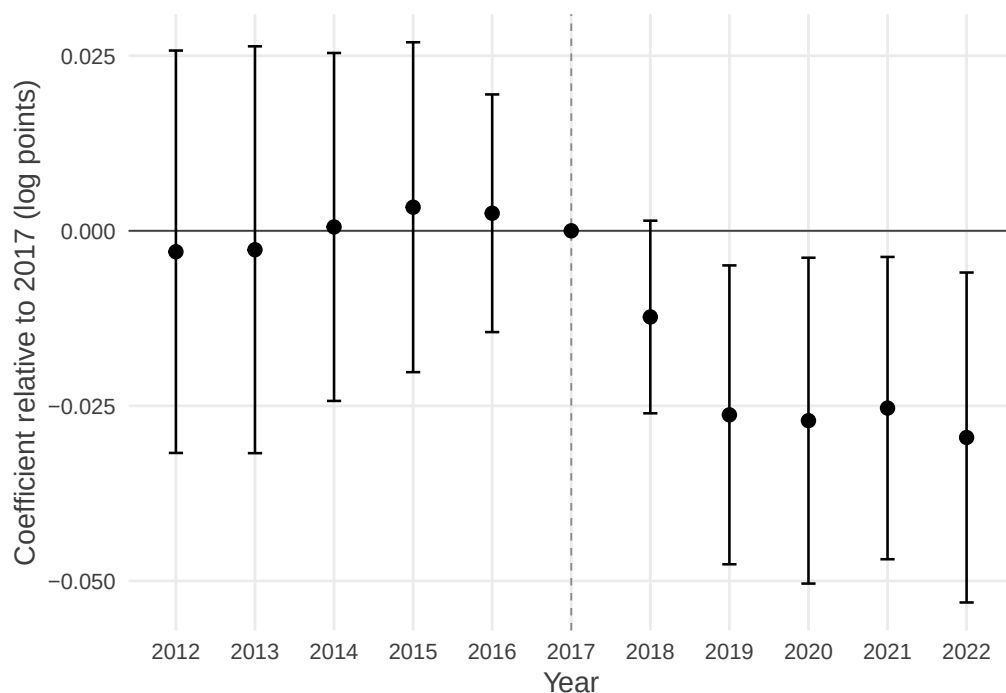
Notes: This figure plots the distribution of the cropland protection pressure index ($Pressure_c$) across counties within each prefecture. Each box summarizes the distribution of $Pressure_c$ among non-core counties within a prefecture: the box represents the interquartile range (25th–75th percentiles), and the thick horizontal line indicates the median. Dots denote individual counties.

Figure B.6: Dynamics of Fallow Status by Years since Reclamation



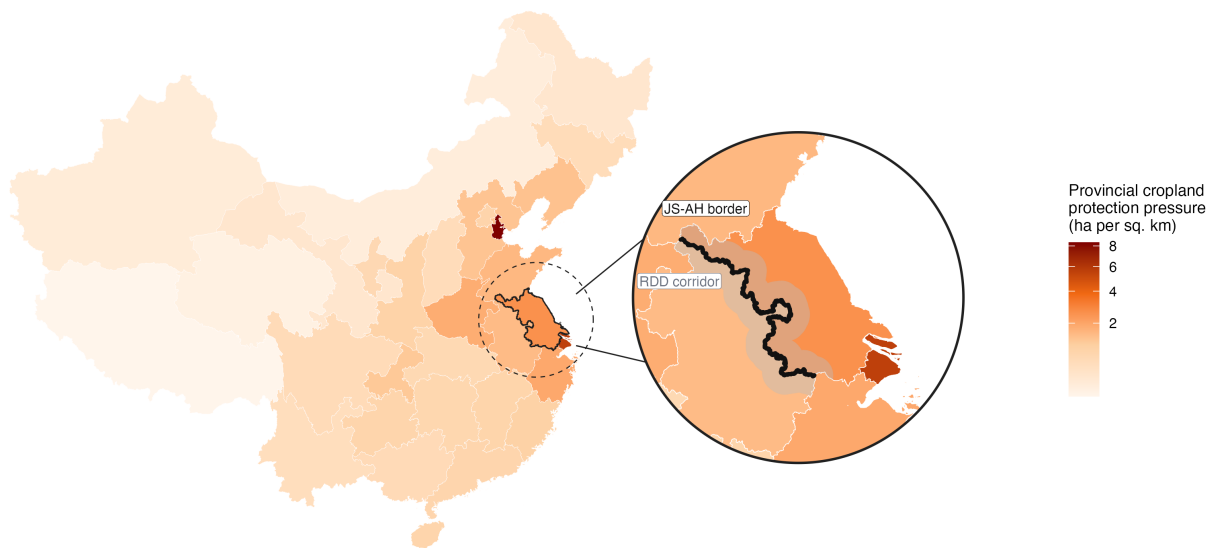
Notes: The figure plots coefficients from Model 3 interacted with years since reclamation, with 95 percent confidence intervals reported.

Figure B.7: Effect of Cropland Protection Pressure on Crop Output per Cropland Hectare



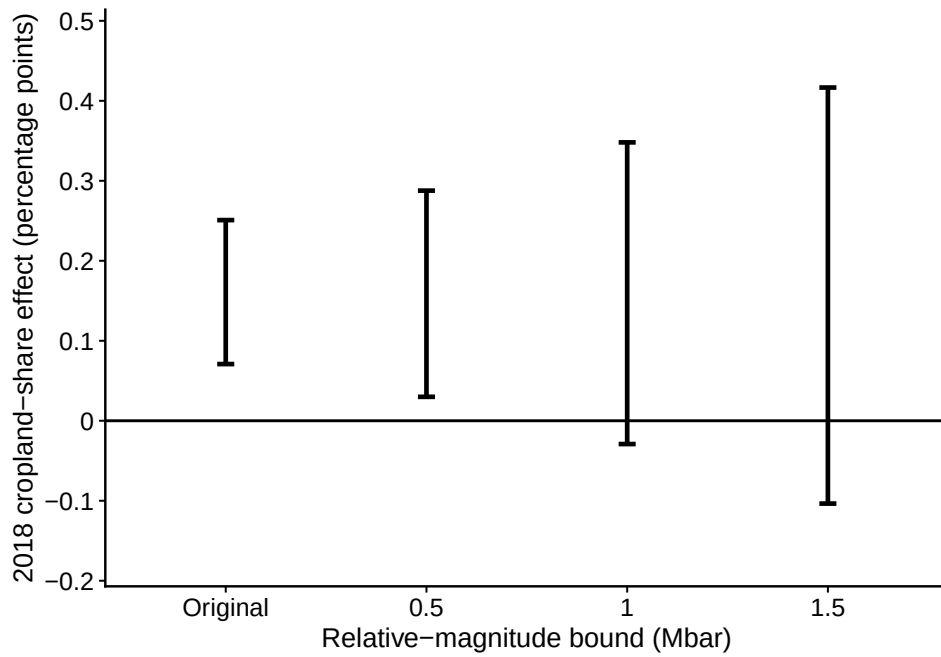
Notes: The figure plots coefficients on interactions between the county-level cropland protection pressure index and calendar-year indicators in a county-year panel, with 2017 omitted. Cropland protection pressure is standardized to mean zero and standard deviation one across the 64 non-core counties, so each coefficient reports the change associated with a one-standard-deviation increase in pressure relative to 2017. The outcome is the log of total grain, oilseed, and cotton output divided by satellite-measured cropland hectares. The specification includes county and year fixed effects. Vertical bars report 95 percent confidence intervals based on standard errors clustered by county. The sample contains 533 county-year observations from 58 counties. A joint test of the 2012–2016 coefficients yields $p = 0.741$.

Figure B.8: Provincial Cropland Protection Pressure at the Border between Jiangsu and Anhui



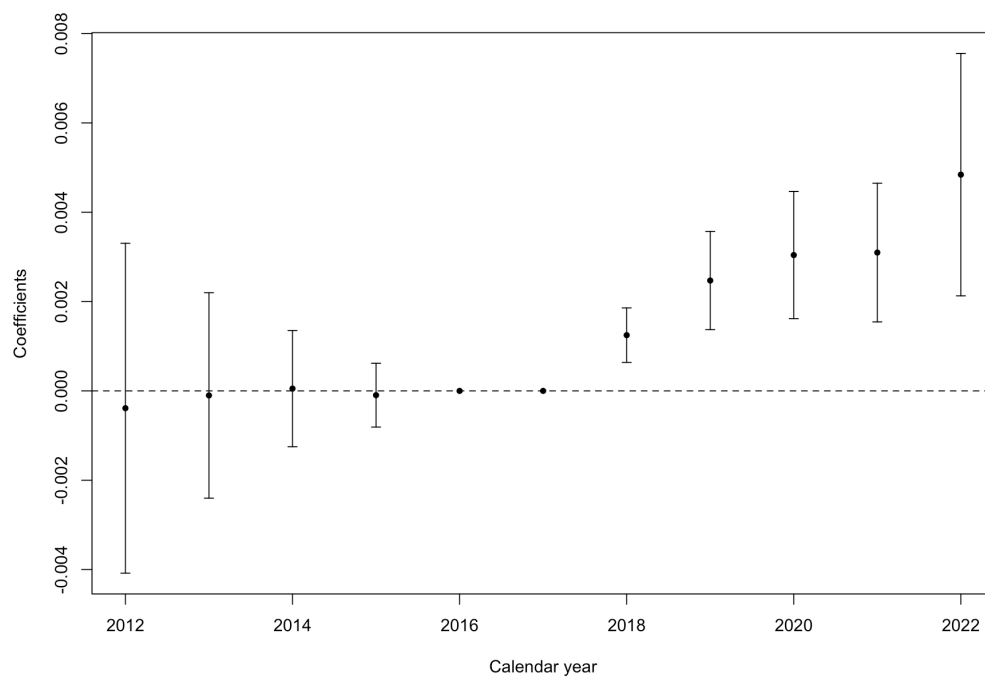
Notes: The national map shades mainland province-level units by provincial cropland protection pressure, measured as hectares of assigned cropland compensation per square kilometer of land area. Darker shades indicate greater pressure. The inset enlarges the region around Jiangsu and marks the spatial RDD corridor along its border with Anhui. Jiangsu faces greater provincial cropland protection pressure than Anhui, while Shanghai faces greater pressure than Jiangsu.

Figure B.9: Rambachan–Roth Bounds for the Cropland-Share Event Study



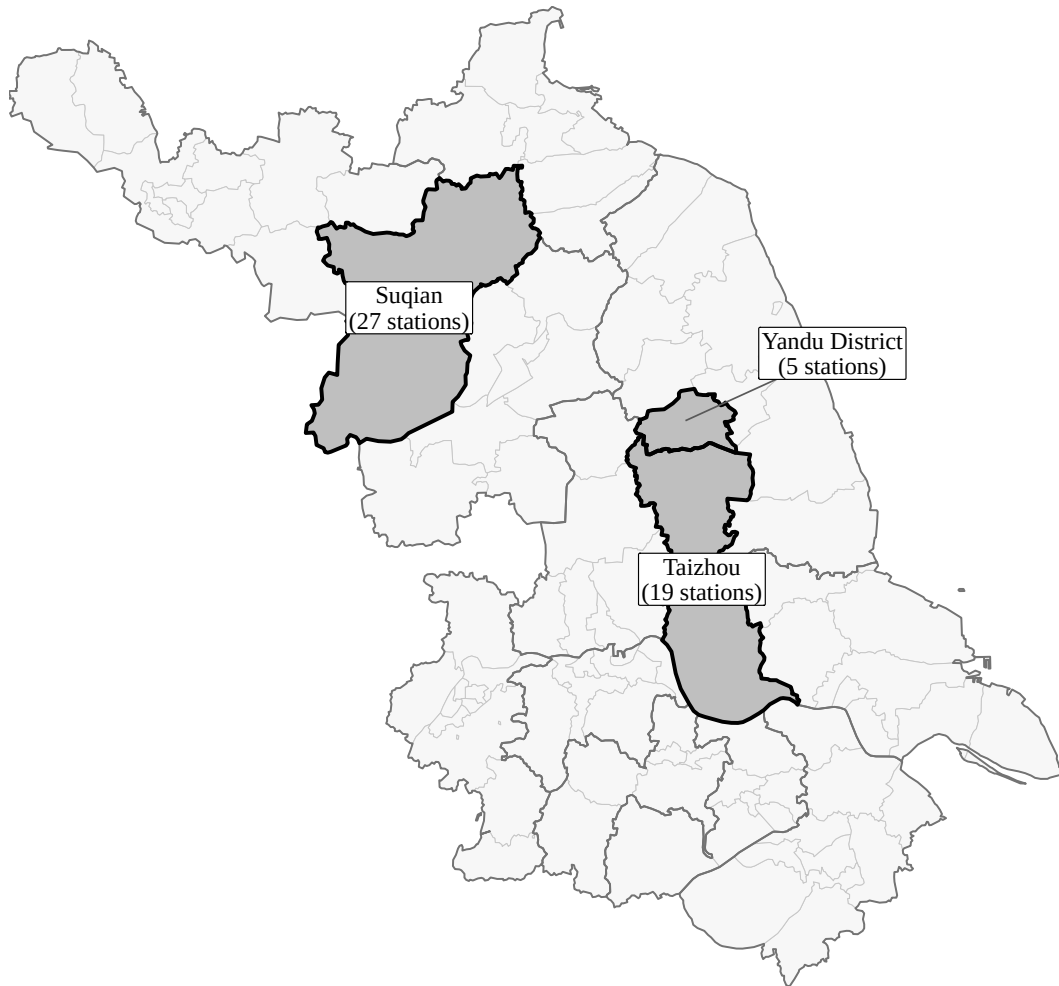
Notes: The figure reports 95 percent confidence intervals for the coefficient on standardized county cropland protection pressure interacted with the 2018 indicator in the event-study specification, relative to the omitted 2017 period. “Original” reports the conventional confidence interval based on standard errors clustered by county. The remaining bars report robust confidence intervals at the indicated values of $\bar{M}$ under the relative-magnitude restriction of Rambachan and Roth (2023). Cropland protection pressure is standardized to mean zero and standard deviation one across the 64 non-core counties, so each interval represents the effect of a one-standard-deviation increase in pressure. The outcome is grid-level cropland share, measured in percentage points. The specification includes grid-cell and year fixed effects and uses 3,743,982 grid-year observations. The bound $\bar{M}$ limits changes in post-treatment deviations from parallel trends relative to the largest corresponding pre-treatment change. The lower endpoint remains above zero at $\bar{M} = 0.75$ and includes zero at $\bar{M} = 0.80$, placing the break value in $(0.75, 0.80]$.

Figure B.10: Cropland-Share Event Study with County-Specific Linear Trends



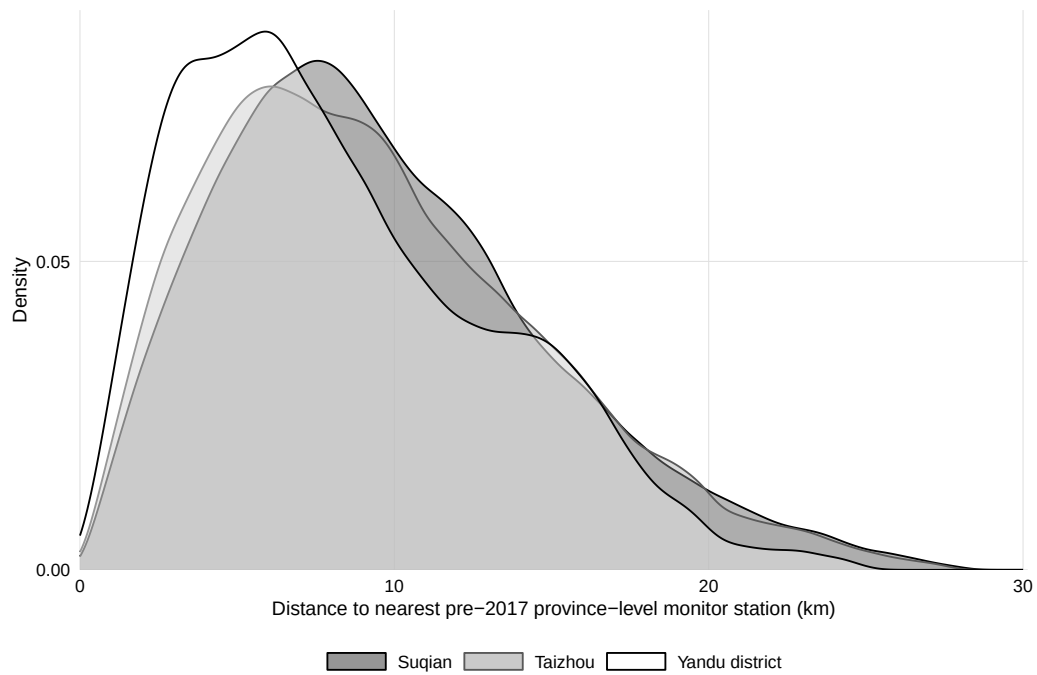
Notes: The specification is identical to Figure 5, except that it adds county-specific linear calendar-year trends.

Figure B.11: Monitoring-Station Sample Areas in Jiangsu



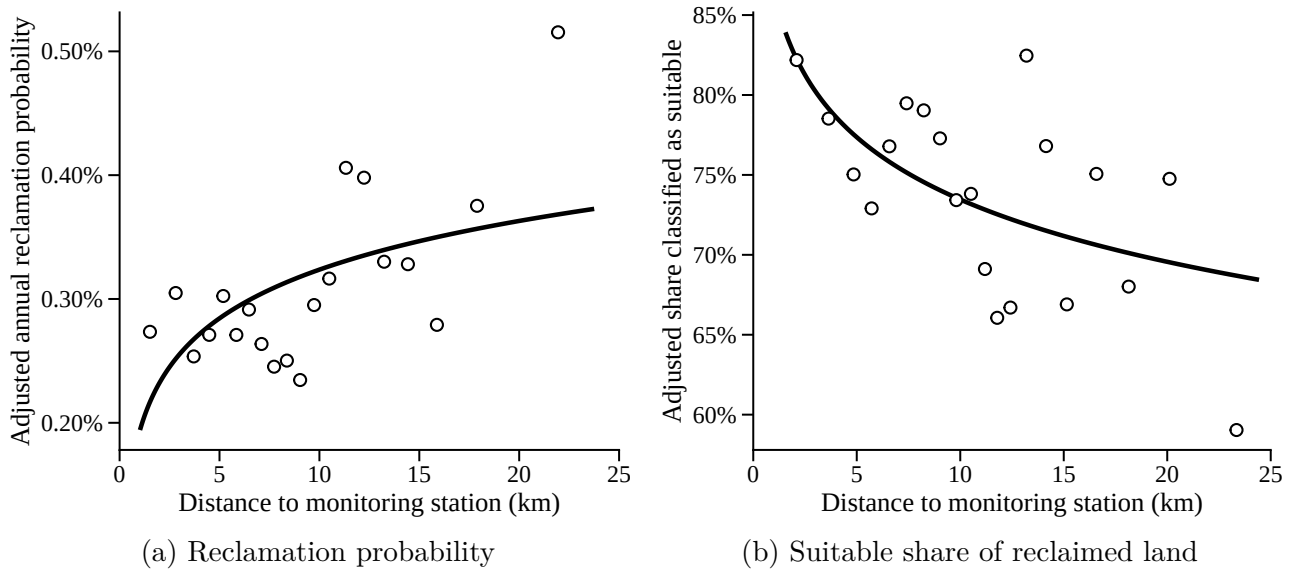
Notes: Shaded areas identify the jurisdictions covered by the monitoring-station sample used in Table 5: Suqian Prefecture, Taizhou Prefecture, and Yandu District of Yancheng Prefecture. Labels report the number of pre-2017 province-level cropland-quality monitoring stations retained in each area: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Thin lines show county boundaries, and thicker lines show prefecture boundaries.

Figure B.12: Distance Distribution in the 100m Monitor-Station Sample



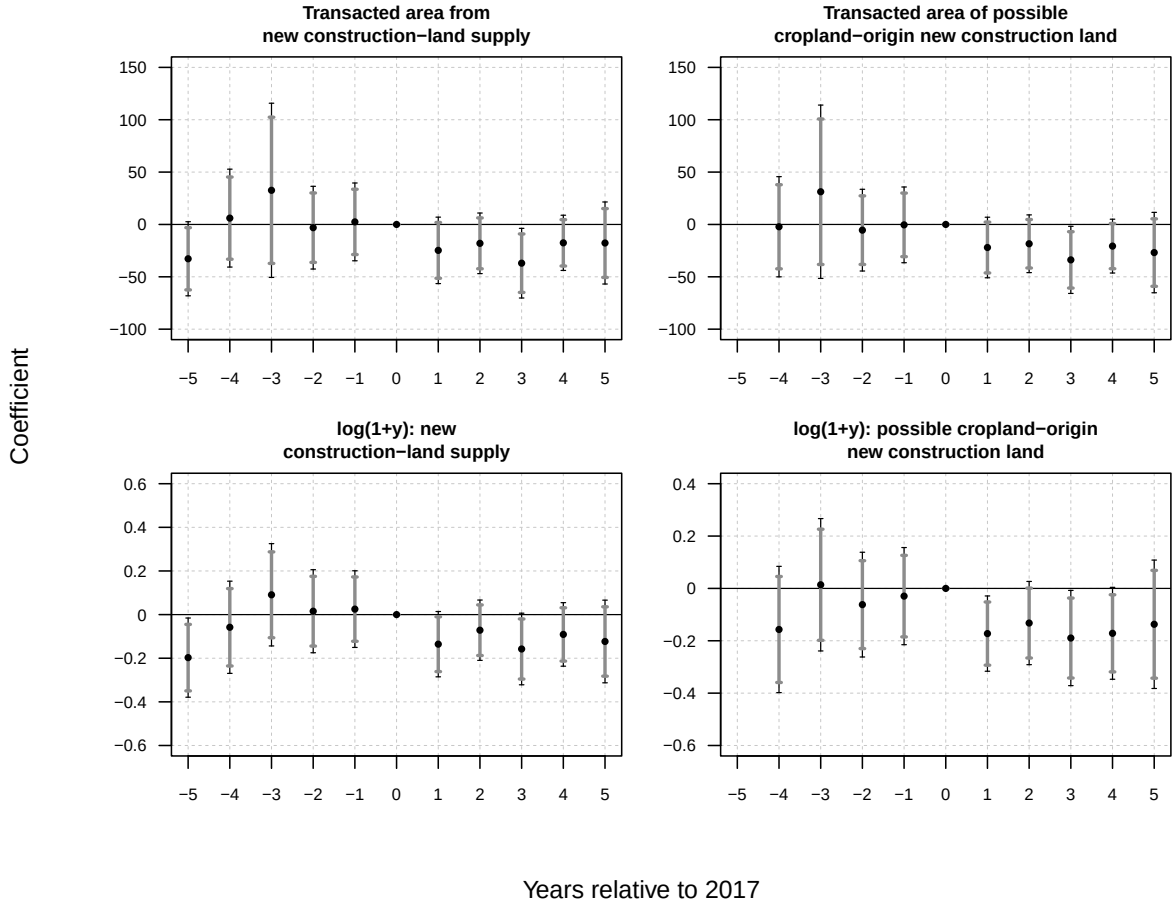
Notes: This figure plots the distribution of distances from 100-meter grid-cell centroids to the nearest pre-2017 province-level cropland-quality monitoring station. The station set matches Table 5: province-level stations in Suqian, Taizhou, and Yandu district. The public reports provide 51 pre-2017 province-level stations: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Across the 1.55 million 100-meter grids in this subsample, the mean distance to the nearest province-level station is 9.47 km, the median is 8.70 km, and the interquartile range is 5.52–12.71 km. Distances are measured in kilometers.

Figure B.13: Monitoring-Station Distance and Post-Policy Reclamation Choices

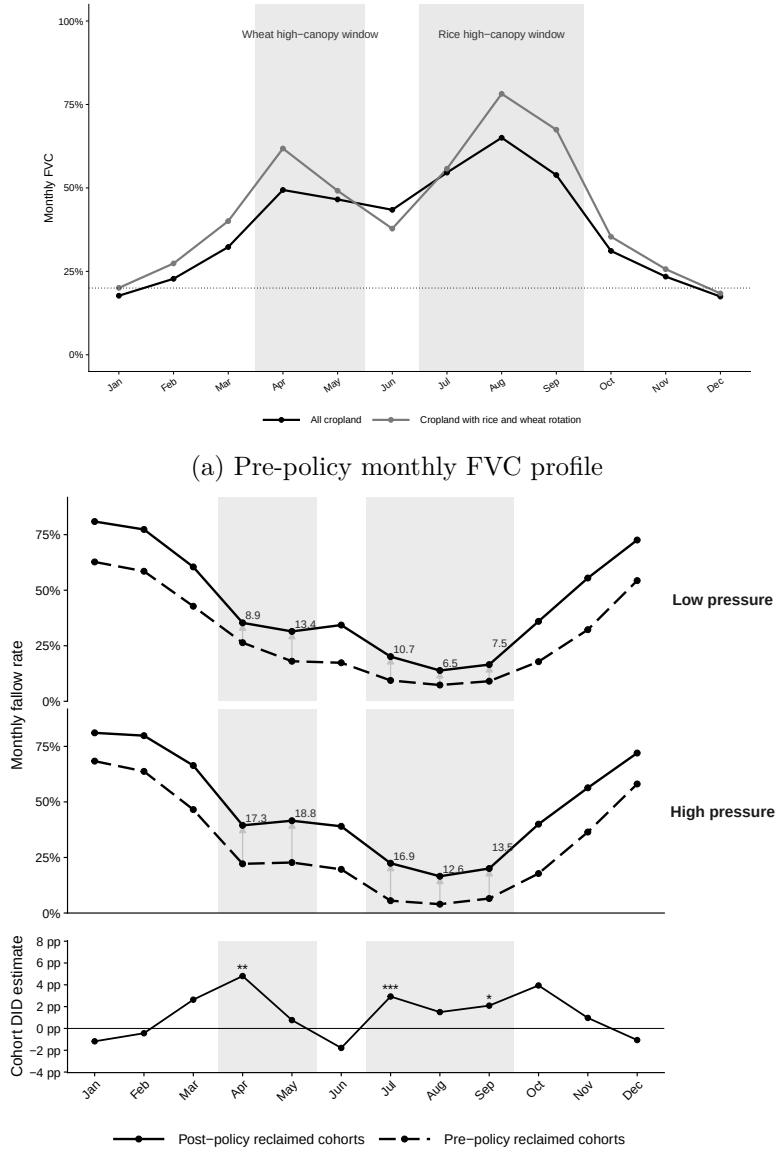


Notes: The horizontal axis measures distance to the nearest cropland-quality monitoring station. Panel (a) uses all 100-meter grid cells and plots annual reclamation probability during 2018–2022. Panel (b) uses grid cells reclaimed during 2018–2022 and plots whether soil is suitable according to the 1995 China Soil Map. The vertical axis reports outcomes residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county fixed effects; panel (b) also includes reclamation-cohort fixed effects. Circles show means within 20 equal-count distance bins, and solid lines show fitted gradients. Standard errors are clustered by parent 500-meter grid. The coefficients on monitoring proximity, $-\log(d_i)$, are -0.0567 in panel (a) and 5.624 in panel (b), expressed in percentage points; both $p < 0.001$.

Figure B.14: Event-Study Estimates for Construction-Land Transaction Supply



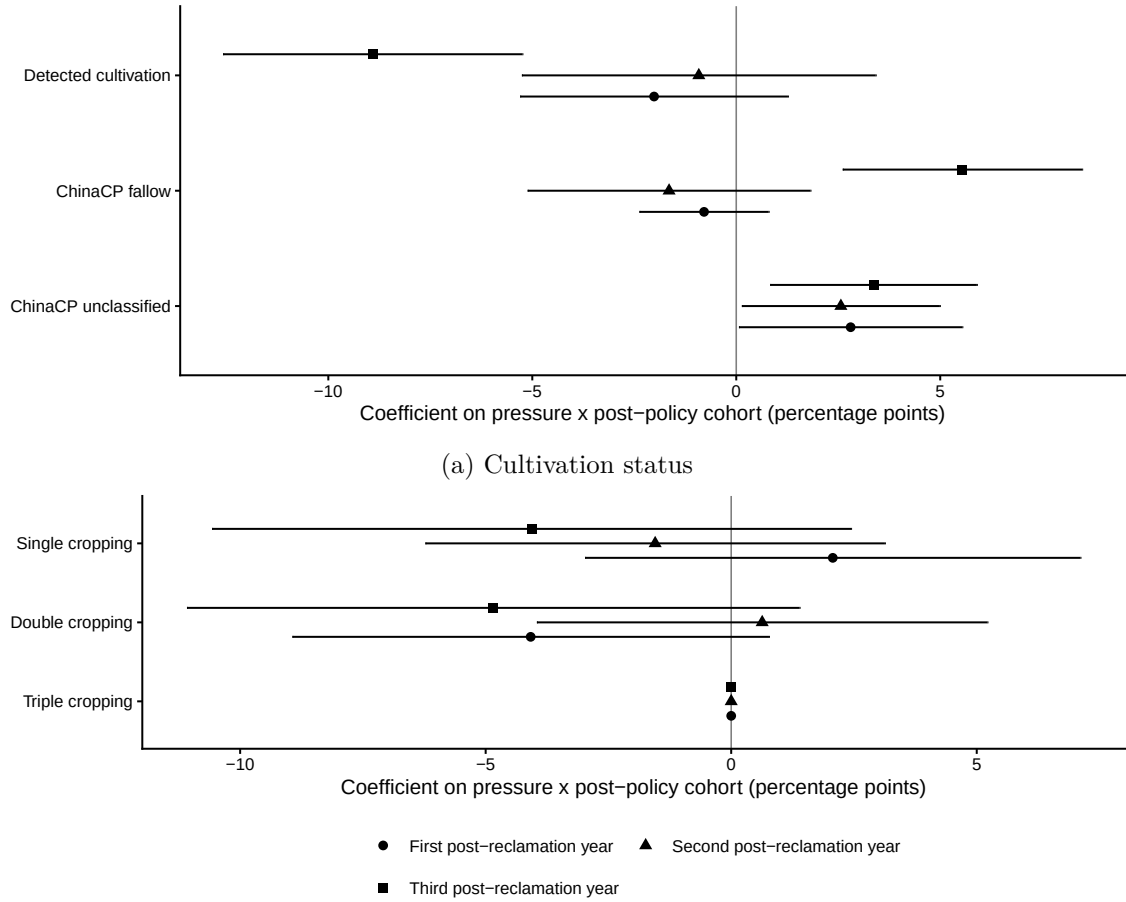
Notes: This figure plots county-level event-study estimates for realized primary-market construction-land transaction records in Jiangsu from 2012 to 2022. The estimating equation interacts county-level cropland protection pressure with year indicators, omitting 2017, and includes county and year fixed effects. The top panels use area outcomes in hectares; the bottom panels use $\log(1 + y)$, where y is the corresponding hectare outcome. The left panels restrict to transacted area from new construction-land supply. The right panels further restrict to transactions in possible cropland-origin new construction land, defined as new-supply transactions whose centroid falls in a 500 m grid cell with any cropland cover in the previous-year land-cover data. Thin bars denote 95% confidence intervals, corresponding to the 5% level; thick bars denote 90% confidence intervals, corresponding to the 10% level. Standard errors are clustered by county.



(b) Monthly fallow-rate profiles and cohort DID estimates

Figure B.15: Seasonal FVC Benchmarks and Monthly Fallow-Rate Effects

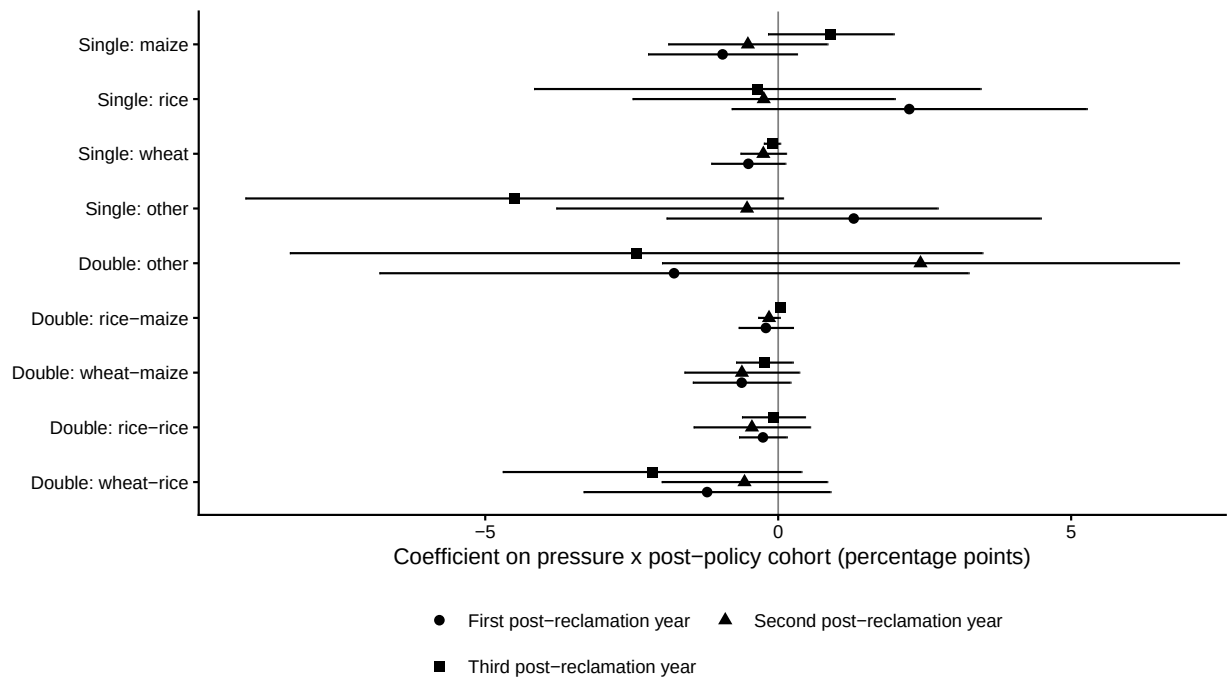
Notes: Panel A plots mean monthly FVC for all cropland in 2012–2017 and for cropland with rice and wheat rotation in 2015–2017, the years for which crop-pattern data are available. The gray shaded regions mark crop-specific high canopy windows: April–May for winter wheat and July–September for rice. The dotted horizontal line marks $FVC = 0.20$, the threshold used to define monthly fallow. Panel B uses newly reclaimed grids observed in the first four years after reclamation, $s \in \{1, 2, 3, 4\}$. The upper two plots report mean monthly fallow rates for pre-policy and post-policy reclaimed cohorts separately in low- and high-pressure counties; high-pressure counties are those above the sample county median of the pressure index. Light-gray arrows connect the pre-policy and post-policy values in the high canopy windows, and the adjacent labels report the increase in percentage points. The lower plot reports month-specific cohort DID coefficients on pressure by post-policy cohort. The cohort DID specifications include county, reclamation-cohort, and years-since-reclamation fixed effects; standard errors are clustered by prefecture. Monthly estimation N ranges from 3,925 to 4,004. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.



(b) Decomposition of detected cultivation

Figure B.16: Crop Patterns among Newly Reclaimed Pixels

Notes: Panel A partitions newly converted CLCD pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified; Panel B decomposes detected cultivation into mutually exclusive single-, double-, and triple-cropping shares. Newly converted pixels are non-cropland in $t - 1$ and remain cropland through $t + 2$, subject to right censoring. ChinaCP unclassified means no detected crop-pattern code and is not treated as observed fallow. Points report Model 3 cohort DID estimates separately by post-reclamation year; horizontal lines show 95 percent confidence intervals. Standard errors are clustered by county. The third-year estimate is identified by the 2018 post-policy cohort.



(c) Crop-specific decomposition

Figure B.16: Crop Patterns among Newly Reclaimed Pixels (continued)

Notes: Panel C decomposes the single- and double-cropping shares in Panel B into nine mutually exclusive crop-specific ChinaCP classes. Triple cropping remains an aggregate category because ChinaCP does not identify its constituent crops. Points report the same Model 3 cohort DID estimates and 95 percent confidence intervals as Panels A and B. Within each post-reclamation year, p -values are adjusted across the nine classes using the Benjamini–Hochberg false discovery rate procedure; no adjusted q -value is below 0.10.

C Tables

Table C.1: Additional Data Sources

Data	Source and coverage
Officials' biographies	Publicly available biographies of prefecture- and province-level officials
Government work reports and quarterly GDP	Municipal Government Work Reports and compiled prefecture-level quarterly GDP releases and statistical summaries, 2012–2022
Construction-land transactions	China Land Market Network primary-market transaction records for Jiangsu, 2012–2022
Cropland-quality monitoring stations	Public cropland-quality monitoring reports for Suqian, Taizhou, and Yandu District; 51 province-level stations established before 2017
Cropping patterns	China Cropping Pattern maps (ChinaCP), 500 m annual, 2015–2021
Crop-specific harvested area	ScienceDB gridded maps for 16 crops, 1 km, five-year intervals, 1990–2020
Annual wetland classifications	Global Annual Wetland Dataset at 30 m with a Fine Classification System from 2000 to 2022 (GAWD) (Zhang et al., 2024). Annual files are archived as Time-Series Global 30 m Wetland Maps from 2000 to 2022 on Zenodo (Liu and Zhang, 2023). Analysis coverage is 2012 to 2021
Elevation and terrain ruggedness	NASA digital elevation model, 30 m
County agricultural values	China County Statistical Yearbooks
Consumer Price Index	World Bank World Development Indicators, China

Notes: Sources listed in Table 1 are not repeated.

Table C.2: Mapping Soil Types to Suitability Classes

Score	FAO Class	Representative Soil Groups
5 (S1)	Highly suitable	Fluvisols, Anthrosols, alluvial soils
4 (S2)	Moderately suitable	Cambisols, Luvisols, Ferralsols, Andosols
3 (S3)	Marginally suitable	Regosols, Leptosols, Cambic soils, hardpan soils
2 (N1)	Currently unsuitable	Saline soils, sodic soils, waterlogged paddy soils
1 (N2)	Permanently unsuitable	Urban land, marsh soils, water bodies, sandbars

Table C.3: Parallel-Trends Tests for the Cohort Difference-in-Differences Design

	China Soil Map 1995 1(Suitable) (1)	HWSD v2 2023 1(Suitable) (2)	Fallow 1(Fallow) (3)
Pressure $\times$ false post (2014)	-0.019 (0.026)	-0.011 (0.028)	0.063 (0.052)
Pressure $\times$ false post (2015)	-0.003 (0.017)	0.009 (0.021)	0.057 (0.051)
Pressure $\times$ false post (2016)	-0.016 (0.019)	-0.022 (0.024)	0.057 (0.046)
County fixed effects	Yes	Yes	Yes
Reclamation-cohort fixed effects	Yes	Yes	Yes
Number of counties	47	47	49
Num. Obs.	448	448	460
R^2 (2014 cutoff)	0.566	0.538	0.348
R^2 (2015 cutoff)	0.566	0.537	0.347
R^2 (2016 cutoff)	0.566	0.538	0.348

Notes: The sample contains the pre-policy reclamation cohorts 2012, 2014, 2015, and 2016. Each row reports a separate regression that defines false post as an indicator for a reclamation cohort at or after the year shown and interacts it with the raw cropland protection pressure index. Each newly reclaimed grid is observed in its first post-reclamation year, so the realized outcomes fall in calendar years 2013–2017. The China Soil Map indicator equals one when the 1995 suitability score is at least three. The HWSD indicator equals one when the grid’s area-weighted HWSD v2 2023 suitability score is at least three; the two soil columns use the same common soil sample as Table 2. Fallow equals one when fractional vegetation cover is below 0.20. All regressions include county and reclamation-cohort fixed effects. Standard errors clustered by county are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Standardized Pressure and Cropland Share

	Cropland share
2012 ($k = -6$) $\times$ Standardized Pressure	-0.002 (0.001)
2013 ($k = -5$) $\times$ Standardized Pressure	-0.001* (0.001)
2014 ($k = -4$) $\times$ Standardized Pressure	-0.001 (0.001)
2015 ($k = -3$) $\times$ Standardized Pressure	-0.001 (0.000)
2016 ($k = -2$) $\times$ Standardized Pressure	-0.000 (0.000)
2018 ($k = 0$) $\times$ Standardized Pressure	0.002*** (0.000)
2019 ($k = 1$) $\times$ Standardized Pressure	0.003*** (0.001)
2020 ($k = 2$) $\times$ Standardized Pressure	0.004*** (0.001)
2021 ($k = 3$) $\times$ Standardized Pressure	0.004*** (0.001)
2022 ($k = 4$) $\times$ Standardized Pressure	0.007** (0.003)
Fixed effects	Grid, year
Cluster level	County
Num. Obs.	3,743,982
R ²	0.981

Notes: Pressure is standardized among non-core counties. Standard errors are clustered at the county level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.5: Standardized Pressure: Newly Reclaimed Cropland

	China Soil Map 1995				HWSD v2 2023		Fallow	
	CSM score (1)	CSM score (2)	CSM suitable (3)	CSM suitable (4)	HWSD2 suitable (5)	HWSD2 suitable (6)	Fallow (7)	Fallow (8)
Standardized Pressure $\times$ Post	-0.202*** (0.064)	-0.223*** (0.071)	-0.057** (0.022)	-0.063*** (0.023)	-0.049*** (0.018)	-0.057*** (0.021)	0.060*** (0.017)	0.061*** (0.019)
County controls $\times$ year		✓		✓		✓		✓
County & cohort FE	✓	✓	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE							✓	✓
Num. Obs.	939	939	939	939	939	939	3,243	3,221
R ²	0.414	0.417	0.452	0.457	0.397	0.401	0.230	0.229

Notes: Pressure is standardized among non-core counties. CSM denotes the 1995 China Soil Map. Standard errors are clustered at the county level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6: Spatial RD Placebo Tests Assigning the Policy Change to Pre-Policy Years

	(1)	(2)	(3)
<i>Panel A: Cropland-share change</i>			
RD in 2013-2012 change (Jiangsu – Anhui)	-0.002 (0.002)	-0.002 (0.002)	-0.001 (0.002)
RD in 2014-2012 change (Jiangsu – Anhui)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
RD in 2015-2012 change (Jiangsu – Anhui)	-0.001 (0.002)	-0.001 (0.002)	0.000 (0.003)
RD in 2016-2012 change (Jiangsu – Anhui)	0.000 (0.003)	0.001 (0.003)	0.001 (0.003)
Eligible cells	161,770	161,770	161,770
<i>Panel B: Fallow-rate change (ex ante marginal cells)</i>			
RD in 2013-2012 change (Jiangsu – Anhui)	0.004 (0.008)	0.003 (0.009)	0.000 (0.010)
RD in 2014-2012 change (Jiangsu – Anhui)	-0.015 (0.014)	-0.017 (0.015)	-0.025 (0.017)
RD in 2015-2012 change (Jiangsu – Anhui)	0.015 (0.018)	0.012 (0.019)	0.022 (0.016)
RD in 2016-2012 change (Jiangsu – Anhui)	0.006 (0.017)	0.004 (0.018)	0.006 (0.017)
Eligible cells	30,600	30,600	30,600
County-pair indicators	Yes	Yes	Yes
First-reclamation-year indicators	No	No	No
Kernel	Triangle	Epanech.	Uniform

Notes. Each row reports a separate sharp spatial RD falsification estimate at the Jiangsu–Anhui border. Outcomes are changes relative to 2012 for 2013–2016. The falsification window ends before the 2017 policy announcement. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. Panel A uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ± 50 km corridor and is not restricted by FVC availability. Panel B uses cells with complete FVC in 2012–2022 and restricts the sample to cells with 2012–2017 mean cropland share below 0.6. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Parentheses contain robust standard errors clustered by county. Eligible cells are counted before local bandwidth selection. Significance: * 10%, ** 5%, *** 1%.

Table C.7: Spatial RD Falsification Checks Using Alternative Outcomes

	<i>N</i>	(1)	(2)	(3)
<i>Pre policy outcomes</i>				
Mean cropland share, 2012–2016 (Jiangsu – Anhui)	161,770	0.010 (0.018)	0.007 (0.017)	0.009 (0.020)
Mean elevation (m) (Jiangsu – Anhui)	160,565	0.350 (2.615)	0.444 (2.651)	0.578 (2.699)
Mean terrain ruggedness index (m) (Jiangsu – Anhui)	160,565	-0.001 (0.255)	0.004 (0.263)	0.003 (0.277)
1(Suitable) (Jiangsu – Anhui)	161,770	-0.015 (0.029)	-0.016 (0.031)	-0.016 (0.034)
Mean annual precipitation, 1970–2000 (mm) (Jiangsu – Anhui)	161,770	1.289 (3.271)	1.621 (3.310)	1.823 (3.364)
<i>Policy invariant outcomes</i>				
Change in mean annual precipitation (mm) (Jiangsu – Anhui)	161,770	-3.792 (3.650)	-4.021 (3.676)	-3.989 (3.762)
Change in mean annual temperature (°C) (Jiangsu – Anhui)	161,770	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
County-pair indicators		Yes	Yes	Yes
First-reclamation-year indicators		No	No	No
Kernel		Triangle	Epanech.	Uniform

Notes. Each row reports a separate sharp spatial RD falsification estimate at the Jiangsu–Anhui border. Coefficients are Jiangsu-minus-Anhui discontinuities. Mean cropland share is measured over 2012–2016. 1(Suitable) equals one when the area-weighted 1995 China Soil Map suitability score is at least 3. Mean annual precipitation is the sum of the 12 WorldClim 2.1 monthly normals for 1970–2000, bilinearly interpolated at each grid centroid. These first five rows are pre policy outcomes. Changes in mean annual precipitation and temperature are the 2018–2022 average minus the 2012–2017 average from TerraClimate v1.1. Mean annual temperature is the day-weighted mean of monthly $(T_{\max} + T_{\min})/2$. These final two rows are policy invariant outcomes because the weather measures should not respond to the 2017 enforcement tightening. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. The sample uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ± 50 km corridor. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Standard errors are clustered at the county level and reported in parentheses. *N* is the outcome-specific complete-case count before local bandwidth selection. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.8: Cropland-Share and Fallow-Rate Discontinuities Across Provincial Borders

	Main comparison			Robustness comparison		
	Jiangsu–Anhui			Jiangsu–Shanghai		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Δ Cropland share</i>						
RD in Δ cropland share (Jiangsu – comparison side)	0.006** (0.003)	0.006** (0.003)	0.007** (0.003)	-0.027* (0.014)	-0.027* (0.015)	-0.029** (0.012)
<i>Panel B: Δ Fallow rate</i>						
RD in Δ fallow rate (Jiangsu – comparison side)	0.037*** (0.010)	0.038*** (0.011)	0.037*** (0.014)	-0.256*** (0.035)	-0.266*** (0.031)	-0.266*** (0.042)
<i>Panel C: Δ Fallow rate (excluding high pre-policy built-up share)</i>						
RD in Δ fallow rate (Jiangsu – comparison side)	0.037*** (0.009)	0.034*** (0.011)	0.039*** (0.011)	-0.205*** (0.025)	-0.206*** (0.024)	-0.214*** (0.019)
<i>Panel D: Δ Fallow rate (excluding high pre- or post-policy built-up share)</i>						
RD in Δ fallow rate (Jiangsu – comparison side)	0.032*** (0.009)	0.030** (0.012)	0.037*** (0.011)	-0.216*** (0.022)	-0.212*** (0.021)	-0.216*** (0.018)
County-pair indicators	Yes	Yes	Yes	Yes	Yes	Yes
First-reclamation-year indicators	No	No	No	No	No	No
Kernel	Triangle	Epanech.	Uniform	Triangle	Epanech.	Uniform

Notes. Columns (1)–(3) compare Jiangsu with Anhui. Columns (4)–(6) compare Jiangsu with Shanghai as a robustness check; Shanghai faces greater provincial cropland protection pressure than Jiangsu. The samples contain cells in matched cross-border county pairs within the ± 50 km corridor; the Jiangsu–Anhui samples also exclude mountain cells. Panel A uses all eligible cropland cells and is not restricted by FVC availability. Its outcome is the cell’s mean cropland share in 2018–2022 minus its mean in 2012–2017. Panel B uses cells with complete FVC in both periods and restricts the sample to ex ante marginal cells with 2012–2017 mean cropland share below 0.6. Its outcome is the change in mean fallow rate between 2012–2017 and 2018–2022. Panels C and D address the possibility that urban expansion lowers FVC on ex ante marginal cells independently of agricultural use. Panel C excludes cells whose mean built-up share in 2012–2017 exceeds 0.8. Panel D excludes cells whose mean built-up share exceeds 0.8 in either 2012–2017 or 2018–2022. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table C.9: Cohort Difference-in-Differences in Post-Reclamation Fallow Rate at the Jiangsu–Anhui Border

	Fallow					
	Pooled		By years since reclamation s			
	(1)	(2)	$s = 1$	$s = 2$	$s = 3$	$s = 4$
Jiangsu $\times$ Post	0.157** (0.068)	0.138 (0.085)	0.022 (0.093)	0.129 (0.097)	0.221* (0.115)	0.362*** (0.107)
Control mean	0.068	0.068	0.099	0.055	0.059	0.059
Cross-border county-pair FE		✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE	✓	✓				
Number of counties	20	20	20	20	20	19
Num. Obs.	1,797	1,797	541	492	433	331
R ²	0.195	0.254	0.298	0.235	0.297	0.316

Notes: Each column reports the cohort-difference-in-differences coefficient on $\text{Jiangsu}_i \times \text{Post}_i$ in a regression of the post-reclamation fallow indicator $\mathbf{1}\{\text{FVC}_{i,t_0(i)+s} < 0.20\}$ on Jiangsu_i , $\text{Post}_i = \mathbf{1}\{\text{cohort}_i \geq 2018\}$, and their interaction. The unit of observation is a 500 m grid cell–years-since-reclamation pair (i, s) , with $s = 1, \dots, 4$ years after the cell’s first reclamation event $t_0(i)$. The sample uses the corrected FVC-complete matched-pair corridor underlying Table C.8 Panels B–D before their marginal-cell restrictions: cells in the ± 50 km Jiangsu–Anhui corridor inside cross-border county pairs, excluding mountain cells and requiring non-missing pre- and post-policy FVC. The sample is further restricted to cells reclaimed at least once during 2013–2022. Columns (1)–(2) report the pooled cohort DiD coefficient, with cohort and years-since-reclamation fixed effects in (1) and adding cross-border county-pair fixed effects in (2). Columns (3)–(6) split the sample by years since reclamation, $s = 1, \dots, 4$, with cross-border county-pair and cohort fixed effects. The cohort fixed effects absorb the pre- vs. post-2017 cohort trend, so the interaction coefficient identifies the difference between the post-2017 Jiangsu fallow excess and the corresponding pre-2017 cross-side gap. The Post main effect is mechanically absorbed by the cohort fixed effect and is therefore omitted. Standard errors are clustered at the county level and reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.10: Robustness: Alternative Definitions of Land-Pressure Exposure

	(1) Cropland Share	(2) $\mathbf{1}(\text{Suitable})$	(3) Fallow
Panel A: All Land Types as Available Land			
Pressure $\times$ Post	0.006*** (0.002)	−0.074*** (0.023)	0.078*** (0.026)
Panel B: Alternative Definition of Core-Urban Districts			
Alternative Pressure $\times$ Post	0.003** (0.002)	−0.048* (0.026)	0.084** (0.032)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Notes: This table examines the robustness of the main results to alternative definitions of cropland protection pressure. In Panel A, *available land* is redefined to include all land types, rather than excluding water bodies and impervious surfaces as in the baseline specification. In Panel B, the classification of *core-urban districts* is tightened by coding only fully core districts as core, while partially core districts are grouped with non-core districts. All specifications otherwise follow the baseline empirical design.

Table C.11: Quantity-Adjusted Historical-Reservoir Decomposition

Panel A. Historical quality prediction, 2002–2017

Predictor	Estimate	Standard error	<i>p</i> -value
Suitable share of remaining reservoir	3.574**	1.424	0.015
Current reclamation / remaining reservoir	0.671	1.749	0.703
County fixed effects		Yes	
Reclamation-cohort fixed effects		Yes	
County-cohort observations		421	
Counties		51	
Within R^2		0.030	
Cluster level		County	

Panel B. Aggregate accounting, 2018–2021

Reference specification	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)	Reference quality
Remaining quality only	9.842	2.429	7.413	24.7	75.3	45.158
Remaining quality + current quantity / remaining reservoir	9.842	2.492	7.350	25.3	74.7	45.095
Remaining quality + current quantity / initial reservoir	9.842	2.605	7.238	26.5	73.5	44.983
Remaining quality + current quantity + interaction	9.842	2.289	7.553	23.3	76.7	45.298

Panel C. Pressure-induced pooled decomposition

Reference specification	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)
Remaining quality only	5.631*** (2.034)	0.325 (0.442)	5.305** (2.213)	5.8	94.2
Remaining quality + current quantity / remaining reservoir	5.631*** (2.034)	0.091 (0.374)	5.539** (2.186)	1.6	98.4
Remaining quality + current quantity / initial reservoir	5.631*** (2.034)	0.043 (0.363)	5.588** (2.179)	0.8	99.2
Remaining quality + current quantity + interaction	5.631*** (2.034)	−0.382 (0.417)	6.013*** (2.141)	−6.8	106.8

Notes: Quality is the percentage of newly reclaimed land classified as suitable, defined as a score of at least 3 in the 1995 soil measure. The main quantity variable is current reclaimed area divided by the factual reservoir remaining at the beginning of the year; both areas are measured in hectares. The interaction specification additionally includes the product of the remaining suitable share and current reclamation intensity. Panel A uses the 2002–2017 historical sample, reclaimed-area weights, county and cohort fixed effects, and county-clustered standard errors. Panel B is a reclaimed-area-weighted accounting exercise covering 92 county-cohort observations in 37 counties. Historical benchmark quality is 47.588 and actual post-policy quality is 37.745. Panel C reports coefficients for a one-standard-deviation increase in cropland-protection pressure from regressions with county and cohort fixed effects, reclaimed-area weights, and county-clustered standard errors. It contains 211 estimable county-cohort observations in 40 counties. Positive values denote quality loss. Total loss equals the frontier component plus the selection wedge. Standard errors condition on the estimated historical-reference coefficients. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Shares outside zero to 100 percent arise when the estimated frontier component is negative and should not be interpreted as conventional proportions.

Table C.12: Land Quality and Fallowing of Newly Reclaimed Land

	Outcome: Fallow	
	(1) Suitability score (1–5)	(2) 1 (Suitable)
Pressure $\times$ Post	0.099*** (0.030)	0.088*** (0.025)
Land quality $\times$ Pressure $\times$ Post	−0.010** (0.005)	−0.037** (0.018)
County, Cohort & Year FE	✓	✓
Num. Counties	52	52
Num. Obs.	3,185	3,185
R ²	0.229	0.229

Notes: This table reports cohort DID estimates using newly reclaimed cropland data. The dependent variable equals one if a newly reclaimed grid cell is fallow in a given post-reclamation year. All specifications include county, reclamation-cohort, and years-since-reclamation fixed effects. Standard errors are clustered at the county level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.13: Calibration Inputs for the Agricultural Economic Magnitude

Component	Symbol	No county controls	County trend controls	Source or definition
Pressure scale	$sd(Pressure_c)$	1.104	1.104	Raw county pressure standardized among non-core counties.
Fallow baseline	$\bar{F}$	0.198	0.198	Control mean from Appendix Table C.5; non-core reclaimed-grid observations with $s = 1, \dots, 4$ and pre-2018 cohorts.
Fallow effect	$\hat{\beta}_F$	0.059 (0.017)	0.060 (0.018)	Cohort DiD from Appendix Table C.5; county, cohort, and years-since-reclamation fixed effects; county-clustered s.e. in parentheses; no-controls $N = 3243$, counties = 54; county-trend-controls $N = 3221$, counties = 53.
Quality baseline	$\bar{Q}$	0.491	0.491	Control mean from Appendix Table C.5; non-core first-reclamation observations with pre-2018 cohorts.
Quality effect	$\hat{\beta}_Q$	−0.049 (0.018)	−0.057 (0.021)	Cohort DiD from Appendix Table C.5; county and cohort fixed effects; county-clustered s.e. in parentheses; $N = 939$, counties = 52.
Quality exponent	Ω	0.615	0.615	Calibrated from land and non-land input shares in Adamopoulos et al. (2024).
Primary-industry baseline value	$\bar{V}_{primary}^{ha}$	RMB 49,856	RMB 49,856	Area-weighted primary-industry value added per CLCD cropland hectare, 2012–2016, expressed at the 2017 price level; valid county-years = 316.
Agricultural baseline value	$\bar{V}_{ag}^{ha}$	RMB 31,156	RMB 31,156	Area-weighted agricultural value added per CLCD cropland hectare within the 2012–2016 baseline window, expressed at the 2017 price level; valid observations cover 2013–2016 because the agricultural value-added field is missing in 2012; valid county-years = 232.
Exact loss share	$Loss_{1sd}^{ag} / \bar{V}^{ha}$	13.2%	14.2%	Exact discrete-change formula in equation (8).
Primary-industry exact loss per hectare	$Loss_{1sd}^{ag,primary} / ha$	RMB 6,590	RMB 7,059	Exact loss share multiplied by baseline value per hectare.
Agricultural exact loss per hectare	$Loss_{1sd}^{ag,agVA} / ha$	RMB 4,118	RMB 4,411	Exact loss share multiplied by agricultural value-added baseline value per hectare.
Primary-industry first-order marginal cost	$MC^{ag,primary}$	RMB 6,760	RMB 7,245	Derivative formula in equation (7).
Agricultural first-order marginal cost	$MC^{ag,agVA}$	RMB 4,224	RMB 4,528	Derivative formula in equation (7) multiplied by the agricultural value-added baseline.

Notes: The table lists the numerical inputs used in Table 9. Monetary values are reported in RMB after converting nominal county values to the 2017 price level using the World Bank China CPI. The 2023 suitability measure comes from HWSD v2 and equals one when the weighted suitability score is at least 3.

Table C.14: Construction-Land Expansion Responses to Pressure

	Total transacted construction-land area		Transacted area from new construction-land supply		Transacted area of possible cropland-origin new construction land	
	Whole county (1)	20km outer ring (2)	Whole county (3)	20km outer ring (4)	Whole county (5)	20km outer ring (6)
Pressure $\times$ Post	-0.113** (0.056)	-0.150** (0.058)	-0.134* (0.077)	-0.180** (0.085)	-0.141* (0.078)	-0.154* (0.091)
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
County linear trends	✓	✓	✓	✓	✓	✓
Observations	704	704	704	704	640	640
Counties	64	64	64	64	64	64
R^2	0.538	0.750	0.459	0.672	0.541	0.689
SE cluster	County	County	County	County	County	County

Notes: The unit of observation is a Jiangsu county-year from 2012 to 2022. The sample excludes fully core-urban districts, retaining non-fully-core districts. Outcomes are $\log(1 + y)$, where y is the county-year sum of transaction area in hectares before the log transformation. Total transacted construction-land area sums all realized primary-market construction-land transaction records. Transacted area from new construction-land supply restricts to parcels whose source category indicates newly converted construction land. Transacted area of possible cropland-origin new construction land further restricts to those new-construction-source parcels whose centroid lies in a 500m grid cell with cropland cover in the previous year; the 2012 proxy is omitted because 2011 grid-level land cover is unavailable. Within each outcome pair, the first column uses all county transactions and the second uses transactions outside the pre-policy built-up patch and within 20 km of that patch's edge. The outer-ring columns focus on urban expansion at the margin, which typically proceeds outward from the existing built-up patch. The coefficient is Pressure $\times$ Post. All specifications include county and year fixed effects and county-specific linear trends. Standard errors are clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

D Policy and Source Documentation

This section reports the policy documents, government materials, news accounts, and historical reports cited collectively as Appendix D in the main text. Each entry states the institutional fact, mechanism, or numerical benchmark for which the source is used.

D.1 National Policy and Land-Use Rules

- **Cropland occupation–compensation.** State Council of the People’s Republic of China (1998), *Implementation Regulations of the Land Administration Law of the People’s Republic of China*, Decree No. 256, effective January 1, 1999. This regulation establishes the national occupation–compensation requirement and the one-for-one replacement rule.
- **National land-use plan.** State Council of the People’s Republic of China (2006), *Outline of the National Overall Land Use Plan (2006–2020)*. This plan provides the basis for the national cropland-retention target and the hierarchy of land-use controls.
- **The 2017 Opinions.** State Council of the People’s Republic of China (2017), *Opinions on Strengthening Cropland Protection and Improving the Balance of Cropland Occupation and Compensation*, January 9, 2017, State Council Gazette, No. 5. This policy supports the discussion of tighter accountability and performance assessment after 2017.
- **Land-conveyance revenue.** General Office of the State Council (2006), *Notice on Regulating the Administration of Revenue and Expenditure from the Conveyance of State-Owned Land-Use Rights*, Document No. 100, effective January 1, 2007. This notice supports the discussion of land-conveyance receipts as local fiscal revenue and their permitted uses.
- **Construction-land conveyance.** Ministry of Land and Resources (2007), *Provisions on the Assignment of State-Owned Construction Land-Use Rights through Invitation to Bid, Auction, and Listing*, Order No. 39, effective November 1, 2007. These provisions establish the public bid, auction, and listing procedures described for construction land.
- **Rural land contracting.** Standing Committee of the National People’s Congress (2018), *Rural Land Contracting Law of the People’s Republic of China*, amended December 29, 2018 and effective January 1, 2019. This law supports the discussion of household contracting, transferable operating rights, and restrictions on changing agricultural use.
- **Land-use controls.** Standing Committee of the National People’s Congress (2019), *Land Administration Law of the People’s Republic of China*, amended August 26, 2019 and effective January 1, 2020. This law provides the land-use categories, planning requirements, and approval process for converting agricultural land.

D.2 Cropland Quality and Monitoring

- **Cropland-quality concerns.** Guanghai Wang (2023), *Earnestly Strengthen Cropland Protection and Reform to Improve the Cropland Occupation–Compensation Balance System*, *People’s Daily*, October 10, 2023, p. 10. This account provides evidence on low-quality reclamation, ecological damage, fallowing, and the later loss of newly created cropland.
- **High-standard farmland construction standard.** State Administration for Market Regulation and Standardization Administration of China (2022), *Well-facilitated Farmland Construction—General Rules*, GB/T 30600–2022, issued March 9, 2022 and effective October 1, 2022. This national standard specifies plot improvement, field-road, irrigation, drainage, and other infrastructure requirements for mechanized and large-scale farming.
- **High-standard farmland upgrading.** Ministry of Agriculture and Rural Affairs (2022), *Guiding Opinions on Promoting the Reconstruction and Upgrading of High-Standard Farmland*, Nong Jian Fa [2022] No. 5, November 9, 2022. This guidance calls for appropriately consolidating and leveling plots and improving field roads to support mechanized operations.
- **National monitoring rules.** Ministry of Agriculture (2016), *Measures for the Survey, Monitoring, and Evaluation of Cropland Quality*, issued June 21, 2016 and effective August 1, 2016. These rules require monitoring sites to represent local soil types and cropping systems.
- **Jiangsu monitoring guidance.** Department of Agriculture and Rural Affairs of Jiangsu Province (2021), *Opinions on Effectively Strengthening Cropland-Quality Monitoring and Evaluation in Jiangsu Province*, Su Nong Ye [2021] No. 37. This guidance supports the description of Jiangsu’s fixed monitoring network and minimum coverage requirements.
- **Nantong implementation.** Nantong Municipal Bureau of Agriculture and Rural Affairs (2024), *Notice on Further Strengthening Cropland-Quality Monitoring and Evaluation*, Tong Nong Fa [2024] No. 9, January 24, 2024. This local rule motivates the use of distance to a monitoring station as a proxy for exposure to quality oversight.

D.3 Provincial Constraints and Indicator Trading

- **Jiangsu.** General Office of the Jiangsu Provincial People’s Government (2021), *Jiangsu Province’s 14th Five-Year Plan for Natural Resources Protection and Utilization*, Su Zheng Ban Fa [2021] No. 41, August 9, 2021. This plan supports the claim that Jiangsu faces tight land availability and competing development and protection demands.
- **Zhejiang.** Zhejiang Online (2024), *Balancing Cropland Occupation and Compensation: Zhejiang Reorganizes Its Cropland-Protection Strategy*, December 27, 2024. This account describes the difficulty of satisfying compensation obligations within a high-pressure

coastal province.

- **Guangdong.** Department of Land and Resources of Guangdong Province (2018), *Guangdong Territorial Plan (2016–2030)*. This plan provides a further example of land-use conflicts in a high-pressure coastal province.
- **Interim trading measures.** General Office of the Jiangsu Provincial People’s Government (2014), *Interim Measures for Trading Supplementary-Cropland Indicators in Jiangsu Province*, General Office Document No. 10. These measures establish the earlier provincial basis for cross-jurisdiction trading of supplementary-cropland indicators.
- **Online trading rules.** Department of Natural Resources of Jiangsu Province (2023), *Detailed Rules for Online Trading of Provincial Supplementary-Cropland Indicators in Jiangsu Province*, February 2023. These rules establish the current online procedures for provincial indicator trading.
- **Transaction results.** Jiangsu Provincial Real Estate Development Center (2019–2026), *Public Results of Online Supplementary-Cropland Indicator Transactions*, Jiangsu Land Market Network, accessed May 24, 2026. These results provide the transaction-price benchmarks used in the paper: RMB 420 per square meter for paddy indicators and RMB 345 for non-paddy indicators.

D.4 Historical Reclamation-Cost Benchmarks

- Ramzi M. Sansur (1995), *Environment and Development Prospects in the West Bank and Gaza Strip*, United Nations Conference on Trade and Development, UNC-TAD/ECDC/SEU/8. Its estimate of about \$100 per dunum for previously cultivated but abandoned land, compared with \$180–200 for never-cultivated marginal land, supports the use of cultivation history as a proxy for reclamation cost.
- H. H. Wooten and Margaret R. Purcell (1949), *Farm Land Development: Present and Future, by Clearing, Drainage, and Irrigation*, U.S. Department of Agriculture, Bureau of Agricultural Economics, Circular No. 825. Its estimate of \$15–30 per acre for lightly vegetated, recently abandoned fields, compared with \$95–160 for woodland converted to crops, provides a second cost benchmark for that interpretation.